

# Practical Fair Data Exchange without In-Circuit Public-Key Operations

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**Abstract.** Fair data exchange (FDE) lets a seller receive payment if and only if the buyer obtains the committed data, and its most practical realizations reduce the task to verifiable encryption under committed key (VECK) over a Reed–Solomon-coded file with Fiat–Shamir sampling. In the state-of-the-art code-based scheme, however, proving is dominated not by the file but by a size-independent constant: in-circuit ElGamal operations tie the sampled ciphertexts to the commitment, and the resulting elliptic-curve gadgets both inflate the circuit and force the proving system onto a slow two-chain inner curve whose key verification is expensive to settle on-chain. We eliminate these ciphertexts—and with them every in-circuit public-key operation. In our construction the polynomial commitment itself certifies sample–file consistency: a KZG subset-consistency check and a single evaluation opening, exposed only as a group element, tie the sample to the committed file, while a lightweight commit-and-prove SNARK certifies only field-arithmetic masking and interpolation relations. We prove security in the algebraic group model and random oracle model under discrete-logarithm-type computational assumptions, dropping the DDH and DCR assumptions of prior VECK schemes. The circuit shrinks by about  $20\times$  and the prover runs directly on BLS12-381: for  $2^{17}$ -element files, cryptographic proof generation drops from 21.1 s to 0.945 s at sample size 512, while cryptographic verification depends only on that sample size. The key stays on a precompile-friendly curve, where its scalar multiplication costs 12,000 gas instead of 0.9–5.9 million for the non-precompiled alternatives.

**Keywords:** Fair data exchange · Verifiable encryption · zk-SNARKs · Reed–Solomon codes

## 1 Introduction

Fair data exchange (FDE) enables a seller to receive payment if and only if a buyer obtains the purchased data. Recent blockchain-based FDE protocols reduce this task to verifiable encryption under committed key (VECK): the

seller publishes ciphertexts and a succinct proof that they are consistent with a public commitment to the file, and a smart contract releases payment only when the matching decryption key is revealed [5].

The main route to practicality, and the focus of this paper, is code-based FDE [7]. The committed file is interpreted as evaluations of a low-degree polynomial and extended to a Reed–Solomon codeword. The verifier audits only a Fiat–Shamir-derived sample, while the redundancy guarantees recovery whenever the number of corrupt symbols remains within the decoding radius. This led first to  $\text{VECK}_{\text{EL}}^+$ , with  $\Theta(\lambda)$ -sized verification, and then to  $\text{VECK}_{\text{EL}}^*$ , which avoids full-file public-key encryption by masking the file with a hash stream and using public-key operations only on sampled positions.

$\text{VECK}_{\text{EL}}^*$  sharply reduced proving time, verification, and communication over earlier VECK constructions. Even so, widely traded files—a full-HD JPEG ( $\approx 500$  KB), an MP3 ( $\approx 3.4$  MB per 3.5 minutes)—amount to at most  $\approx 2^{17}$  field elements. There,  $\text{VECK}_{\text{EL}}^*$ ’s proving time is dominated not by the  $O(n)$  file term but by a size-independent constant arising from the in-circuit ElGamal operations on the sampled positions. Being size-independent, this constant burdens every exchange, so eliminating it is the key to reducing prover cost across all file sizes.

This constant is expensive for a structural reason. To tie its sampled ciphertexts to the commitment,  $\text{VECK}_{\text{EL}}^*$  proves that each sampled ciphertext is well formed under the secret key, together with the key relation  $\text{vk} = h^{\text{sk}}$ . Both are statements about group elements, so the circuit must carry elliptic-curve gadgets. These gadgets have two costs: they dominate proving time, and—because the in-circuit ciphertexts live in the pairing group of the KZG commitment—they force the proving system onto a two-chain inner curve that is both slower to prove on and, when a key must later be checked on-chain, expensive to settle. Our starting point is that these ciphertexts, and with them all curve arithmetic, need not appear in the circuit at all.

We therefore break from the public-key encryption primitives—ElGamal and Paillier—that prior VECK schemes rely on, replacing the sampled ciphertexts with a native polynomial-commitment argument. The prover derives the sampled set, builds a low-degree polynomial that agrees with the committed file on those positions, certifies this agreement with a KZG subset-consistency check, and opens the polynomial at a Fiat–Shamir challenge point—exposing the opening only as a group element  $U_\alpha = g_1^{u_\alpha}$ . A commit-and-prove SNARK is then left with only the field-arithmetic masking and interpolation relations, and CP-links bind the hidden scalars to  $U_\alpha$  and  $\text{vk} = h^{\text{sk}}$ , so the circuit performs no elliptic-curve operation.

This removal has three effects. First, the circuit shrinks by about  $20\times$ , the direct source of the prover speedup. Second, with no curve arithmetic left, the proving system is no longer tied to a pairing-friendly inner curve: it can use BLS12-381 directly, which speeds proving further and—because the key-verification group is now free to be precompile-friendly—cuts the on-chain settlement cost by orders of magnitude. Third, exposing the opening as a group

element does double duty: beyond linking, it also carries the soundness extraction and the zero-knowledge simulation under a universal KZG CRS whose trapdoor is unknown to the prover. The ingredients are standard—KZG subset proofs, a CP-SNARK with LegoSNARK-style linking, and Fiat–Shamir sampling.

*Our contributions.*

- A code-based FDE construction without in-circuit public-key operations: a KZG subset-consistency check and a single group-encoded opening tie the sample to the committed file, while a commit-and-prove SNARK certifies only the masking and interpolation relations.
- A security analysis in the AGM and ROM, with full soundness and zero-knowledge bounds. The group-encoded opening makes the composition simulatable under a universal KZG CRS, and hash-based masking keeps all assumptions discrete-logarithm type (no DDH or DCR).
- An implementation with large prover-side gains over  $\text{VECK}_{\text{EL}}^*$  ( $21.1 \rightarrow 0.945$  s at  $2^{17}$  elements,  $R = 512$ ), from an  $\approx 20\times$  constraint reduction plus the freedom to use BLS12-381, while preserving sample-size verification.
- An on-chain cost comparison: our key stays on a precompile-friendly curve, costing 12,000 gas for scalar multiplication versus 0.9–5.9 million on the non-precompiled alternatives.
- A degree-enforcement extension closing the malicious-committer case: one extra pairing check, independent of file length, forces  $\deg \phi \leq \ell$ , so the construction also serves a self-committing seller.

## 2 Related Work

*Fair exchange and verifiable encryption.* Classical fair exchange is impossible in full generality without some trusted mechanism [1]; optimistic protocols minimize this trust by contacting a third party only on aborts or disputes [2, 3]. Blockchain-based FDE replaces the dispute path with an on-chain payment condition, as in smart-contract-based exchange systems such as FairSwap [4] and the more recent VECK-based approach of Tas et al. [5]. VECK is closely related to verifiable encryption [6]: the proof must bind encrypted or masked data to a committed witness while hiding the witness before key release. This work keeps the VECK abstraction and focuses on reducing the prover cost of its most practical code-based instantiation.

*Code-based FDE and polynomial commitments.* Prior code-based FDE work uses Reed–Solomon redundancy and Fiat–Shamir sampling to avoid verifying every ciphertext symbol [7].  $\text{VECK}_{\text{EL}}^+$  reduces verification to a sampled check but uses full-file exponential ElGamal [8], while  $\text{VECK}_{\text{EL}}^*$  replaces the full-file public-key layer with hash masking and sampled ElGamal. Our construction reuses only the coding-theoretic sampling analysis; the consistency layer is redesigned, replacing the sampled ElGamal proof with KZG subset consistency and a commit-and-prove SNARK. This situates the construction in the line of polynomial and

vector commitments—from Pedersen [9] to KZG [10] and transparent Reed–Solomon proximity proofs [11]—and of succinct proofs from CS proofs [12] and verifiable computation [13, 14] to SNARKs; commit-and-prove SNARKs supply the interface that keeps scalar openings hidden while binding them to public group elements [15, 18–20].

### 3 Preliminaries

This section recalls the notation and cryptographic background used throughout the paper. We follow notation common to code-based fair data exchange, adopting the conventions of [7] where convenient.

#### 3.1 Notation

Let  $\lambda \in \mathbb{N}$  be the security parameter;  $\mu(\lambda)$  is *negligible* if it is eventually smaller than every inverse polynomial, and  $x \xleftarrow{\$} S$  denotes uniform sampling from a finite set  $S$ .

Let  $G_1, G_2, G_T$  be bilinear groups of prime order  $p$  with scalar field  $\mathbb{F}_p$ , which carries all arithmetic objects of the protocol: file blocks, Reed–Solomon codewords, random-oracle masks, decryption keys, KZG trapdoors and openings, Fiat–Shamir challenge points, and SNARK variables; thus  $\mathbb{F}_p[X]$  is the univariate polynomial ring. Our implementation instantiates  $(G_1, G_2, G_T)$  with BLS12-381, but the construction and security argument rely only on the bilinear-group interface. For an integer  $B \geq 0$ , write  $[B] := \{0, 1, \dots, B\}$ .

A file is represented by evaluations of a polynomial  $\phi(X) \in \mathbb{F}_p[X]$  of degree at most  $\ell$ , identified when convenient with the vector  $(\phi(0), \phi(1), \dots, \phi(\ell))$ . For a finite ordered set  $S \subseteq \mathbb{F}_p$ , the vanishing polynomial, Lagrange basis, and interpolation of  $\phi$  on  $S$  are

$$V_S(X) := \prod_{i \in S} (X - i), \quad L_{i,S}(X) := \prod_{\substack{j \in S \\ j \neq i}} \frac{X - j}{i - j}, \quad \phi_S(X) := \sum_{i \in S} \phi(i) L_{i,S}(X),$$

so that  $\phi_S(i) = \phi(i)$  for all  $i \in S$  and  $\deg(\phi_S) < |S|$ .

#### 3.2 Bilinear Groups and KZG Commitments

We use generators  $g_1 \in G_1$ ,  $g_2 \in G_2$ , a pairing  $e : G_1 \times G_2 \rightarrow G_T$ , and the KZG polynomial commitment scheme [10]. Given a maximum degree  $n$ , KZG setup samples  $\tau \xleftarrow{\$} \mathbb{F}_p$  and publishes  $crs = (\mathcal{G}, \{g_1^{\tau^i}\}_{i=0}^n, \{g_2^{\tau^i}\}_{i=0}^R)$ , where  $\mathcal{G}$  is the bilinear-group description. A polynomial  $\phi$  of degree at most  $n$  is committed as  $C_\phi := g_1^{\phi(\tau)}$ ; to open  $\phi(\alpha) = y$  the prover sends  $\pi_\alpha := g_1^{q_\alpha(\tau)}$  with  $q_\alpha(X) := (\phi(X) - y)/(X - \alpha)$ , and the verifier checks

$$e(C_\phi / g_1^y, g_2) = e(\pi_\alpha, g_2^{\tau - \alpha}).$$

In the security proof we use the standard extractable form: for an algebraic adversary, an accepting KZG transcript fixes a polynomial of degree at most  $n$  and its claimed openings, except with failure probability  $\epsilon_{\text{KZG}}$ . We use this both for ordinary openings and for subset-opening equations whose quotient is certified by  $V_S(\tau)$ , matching the usual algebraic-group/ $q$ -SDH-style extractability analysis.

### 3.3 Commit-and-Prove SNARKs

We use a commit-and-prove SNARK (CP-SNARK) for consistency: its statement says that public commitments contain hidden values satisfying a relation. Following LegoSNARK [15], its witness splits into committed values  $(u_j)$  and auxiliary data  $\omega$ . It proves that public commitments open to  $(u_j)$  satisfying a relation  $\mathcal{R}(x, (u_j), \omega)$ , through **KeyGen**, **Prove**, and **VerProof**. We require a simulated setup and a simulator that produces an accepting proof for every well-formed public statement, including false ones; indistinguishability from honest proofs is required only for true statements.

*The linking gadget.* The component we rely on most directly is the *linking gadget* CLink, the  $\text{CP}_{\text{link}}$  primitive of LegoSNARK [15]: it proves that two commitments under different keys open to the same value, the minimal ingredient that makes a committed scalar interoperable with a group element produced outside the proof system. This statement is a linear-subspace relation  $[x]_1 = [\mathbf{M}]_1 \cdot w$ , so it can be realized by any QA-NIZK for linear subspaces. We use the LegoSNARK link [15], instantiated with Kiltz–Wee [18]. Its AGM knowledge soundness applies to witness-samplable subspace matrices under DL; our two single-base equality links have this form. The gadget is succinct: one multi-exponentiation to prove and a constant number of group elements and pairings to verify.

### 3.4 Fair Data Exchange and VECK

FDE exchanges payment for data by releasing funds only after a valid decryption key appears [5]. At its core is VECK—verifiable encryption under a committed key—which lets a prover convince a verifier that a ciphertext encrypts data consistent with a public commitment, under a secret key whose verification key is already public. A VECK scheme consists of the following algorithms:

- $\text{Gen}(crs) \rightarrow pp$ : generates public parameters;
- $\text{Enc}(F, C_w, w) \rightarrow (vk, sk, ct, \pi)$ : outputs a verification key, a decryption key, ciphertexts, and a proof;
- $\text{Ver}_{ct}(F, C_w, vk, ct, \pi) \rightarrow \{0, 1\}$ : verifies ciphertext consistency;
- $\text{Ver}_{key}(vk, sk) \rightarrow \{0, 1\}$ : verifies that  $sk$  matches  $vk$ ;
- $\text{Dec}(F, sk, ct) \rightarrow v$  or  $\perp$ : decrypts the ciphertext.

We use the correctness, soundness, and pre-key-release zero-knowledge notions of Tas et al.; Appendix A.1 states the games. Here the committed witness is a polynomial  $\phi$ , and  $F$  returns its designated evaluations.

### 3.5 Reed–Solomon Redundancy

We view the committed evaluations not only as the original file  $(\phi(0), \dots, \phi(\ell))$  but as part of a longer Reed–Solomon (RS) codeword  $(\phi(0), \dots, \phi(m))$ ,  $m \geq \ell$ . This redundancy lets the verifier check only a small sampled subset while preserving recoverability of the original polynomial: if the received vector differs from a valid RS codeword in at most  $t \leq \lfloor (m - \ell)/2 \rfloor$  positions, standard RS decoding recovers the unique polynomial of degree at most  $\ell$  [16]. We model this with two black-box procedures for degree bound  $\ell$ , writing  $y = (y_i)_{i \in S}$ :

- $\text{RS.Det}(S, y) \rightarrow \{0, 1\}$ , returning 0 exactly when  $\{y_i\}$  is the evaluation of a degree-at-most- $\ell$  polynomial; and
- $\text{RS.Dec}(S', S, y) \rightarrow \{\phi(i)\}_{i \in S'} \cup \{\perp\}$ , returning the unique bounded-distance decoding, or  $\perp$  on failure.

In decryption, the client unmaskes the received symbols and invokes  $\text{RS.Det}$ ; if no error is detected it outputs the relevant prefix directly, otherwise it runs  $\text{RS.Dec}$ .

### 3.6 Fiat–Shamir Sampling

The verifier checks only a pseudorandom subset of coordinates via the Fiat–Shamir heuristic [17]. Let  $R$  be the sample budget and  $H$  a hash function modeled as a random oracle. Masking, subset sampling, and point sampling use separate domain tags over an injective transcript encoding: field and group elements are fixed-width canonical encodings, and vectors are length-prefixed. We define

$$S_R := \text{DeriveSubset}_H(C_\phi, \text{vk}, \{ct_i\}_{i \in [m]}, m, R)$$

as a deterministic procedure that derives an  $R$ -element subset  $S_R \subseteq [m]$  from the transcript, instantiated by hashing the transcript with an incrementing counter and rejection-sampling distinct candidate indices in  $[m]$ . Because the subset depends on the transcript, the prover must commit to the masked codeword before learning which positions will be audited.

## 4 Model and Assumptions

*Parties and threat model.* We consider the standard VECK-based FDE flow, which separates two roles. A *committer* publishes a commitment  $C_\phi$  to the file—as in Ethereum Danksharding, where a KZG commitment to the blob persists on-chain after the data expires [5]—and a *seller* then produces masked ciphertext symbols and a consistency proof for a buyer and later reveals a decryption key. Our adversary is the seller, modeled as *fully malicious*: it chooses  $(\text{vk}, \text{sk}, ct, \pi)$  freely and may try to obtain payment while sending ciphertexts that do not decode to the committed file (Definition 2). The commitment  $C_\phi$ , by contrast, is a trusted public input, generated honestly as a degree- $\ell$  commitment—the standard VECK soundness model of Tas et al. and Khabbazi et al. [5, 7]. This matches the motivating deployment: in Danksharding the blob has a protocol-fixed length, so its degree bound is enforced structurally by the base layer rather

than merely trusted. The blockchain or payment contract faithfully executes key verification.

The single case a trusted commitment leaves open is a *malicious committer*, most naturally when the seller commits to the file itself: a commitment to a polynomial of degree above  $\ell$  can escape the Reed–Solomon decoding guarantee even when every sampled check passes. We go one step further and close this case as well: Proposition 1 adds a single one-time pairing check—one extra CRS element, one verification, independent of the file length—that forces  $\deg \phi \leq \ell$ , so the same construction serves both an externally trusted commitment and a self-committing seller with no change to the exchange itself.

*Transcript processing versus proof verification.* The full masked codeword is delivered off-chain, and the Fiat–Shamir sample is derived from the fixed ciphertext transcript. Throughout the evaluation, “verification time” refers to the cryptographic verification after the transcript is fixed and the sampled entries are available, so the checking cost depends on the sample size  $R$ , not the file length. We adopt this convention from the prior code-based constructions we compare against [7], which hash the transcript incrementally as it arrives and likewise report an  $R$ -dependent, file-size-independent cost. The one-pass transcript hash and parsing are excluded for all compared schemes; they remain linear in the transmitted codeword and are not part of the verification figures below.

*Sampling parameters.* Let  $m = \lceil \beta(\ell+1) \rceil - 1$ . Write  $q_S$  and  $q_P$  for the numbers of complete transcripts on which an adversary obtains subset and point challenges. Concrete  $\lambda$ -bit sampling soundness requires

$$q_S \left( \frac{\beta+1}{2\beta} \right)^R \leq 2^{-\lambda}, \quad \beta \geq \beta_{\min}(\lambda, R, q_S) := \frac{1}{2^{1-(\lambda+\log_2 q_S)/R} - 1}.$$

The theorem keeps both grinding terms explicit. The protocol also requires  $m+1 \geq R$ . For shorter files one must either increase the effective redundancy to  $\lceil \beta(\ell+1) \rceil \geq R$  (padding the RS domain) or instantiate a smaller  $R$ .

## 5 Construction

We now present our code-based VECK construction (Figure 1). The committed file  $\phi$  of degree at most  $\ell$  is extended to a Reed–Solomon codeword over  $[m]$ ,  $m = \lceil \beta(\ell+1) \rceil - 1$  for a redundancy factor  $\beta > 1$ ; the verifier checks only a Fiat–Shamir-derived sample, while RS detection and decoding guarantee recovery within the decoding radius.

### 5.1 Overview

Prior code-based schemes combine RS redundancy with Fiat–Shamir sampling to reduce client verification to essentially  $\Theta(\lambda)$ ;  $\text{VECK}_{\text{EL}}^*$  further reached near-plaintext communication via hash masking with public-key work on a sampled

**Gen**( $crs$ )  $\rightarrow pp$ : On input  $crs = (\mathcal{G}, \{g_1^{\tau^i}\}_{i=0}^n, \{g_2^{\tau^i}\}_{i=0}^R)$

1. Set  $h := g_1$ .
2.  $crs_z \xleftarrow{\$} \text{zk-CP-SNARK.KeyGen}(g_1, h, R)$ , including the link key.
3. Output  $pp = (crs, crs_z, h)$ .

**Enc**( $F_{[\ell]}, C_\phi, \phi(X)$ )  $\rightarrow (vk, sk, ct, \pi)$ :

1. Sample  $s \xleftarrow{\$} \mathbb{F}_p$ .
2. Set  $sk := s$  and  $vk := h^s \in G_1$ .
3. Compute  $ct_i := \phi(i) + H(sk, i)$  for all  $i \in [m]$   
(on an  $H$ -domain separate from  $\text{DeriveSubset}_H$  and  $\text{DerivePoint}_H$ ).
4. Derive  $S_R := \text{DeriveSubset}_H(C_\phi, vk, \{ct_i\}_{i \in [m]}, m, R)$ .
5. Define  $\phi_{S_R}(X) := \sum_{i \in S_R} \phi(i) L_{i, S_R}(X)$ .
6. Sample  $t_1, t_2 \xleftarrow{\$} \mathbb{F}_p$  and compute  $\tilde{\phi}_{S_R}(X) := \phi_{S_R}(X) + (t_1 + t_2 X) V_{S_R}(X)$ .
7. Compute  $C_{S_R} := \text{Commit}(crs, \tilde{\phi}_{S_R}(X))$ .
8. Compute  $\pi_{S_R} := \text{Commit}\left(crs, \frac{\phi(X) - \tilde{\phi}_{S_R}(X)}{V_{S_R}(X)}\right)$ .
9. Let  $S'_R := S_R \cup \{-1, -2\}$ , where the two auxiliary points are fixed outside  $[m]$ .
10. For each  $i \in \{-1, -2\}$ , compute  $ct_i := \tilde{\phi}_{S_R}(i) + H(sk, i)$ .
11. Compute  $\alpha := \text{DerivePoint}_H(C_\phi, vk, \{ct_i\}_{i \in [m] \cup \{-1, -2\}}, m, R, C_{S_R}, \pi_{S_R}, S_R)$ .
12. Compute  $u_\alpha, \pi_\alpha := \text{KZG.Open}(crs, \alpha, \tilde{\phi}_{S_R}(X))$  and  $U_\alpha := g_1^{u_\alpha}$ .
13. Compute a zk-CP-SNARK proof  $\pi_Z$  for the relation  $R_{\text{mask}}$ , whose public statement **stmt** and witness **wit** are:  

$$\text{stmt} = (h, S'_R, \{ct_i\}_{i \in S'_R}, \{L_{i, S'_R}(\alpha)\}_{i \in S'_R}, U_\alpha, vk), \quad \text{wit} = (u_\alpha, sk, \{x_i\}_{i \in S'_R}),$$
such that  $\forall i \in S'_R : ct_i = x_i + H(sk, i); \quad u_\alpha = \sum_{i \in S'_R} x_i L_{i, S'_R}(\alpha);$   
 $\text{CPLink}(g_1, U_\alpha, u_\alpha) = 1; \quad \text{CPLink}(h, vk, sk) = 1.$
14. Output  $(vk, sk, ct = \{ct_i\}_{i \in [m] \cup \{-1, -2\}}, \pi = (C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha, \pi_Z))$ .

**Ver**<sub>ct</sub>( $F_{[\ell]}, C_\phi, vk, ct, \pi$ )  $\rightarrow 0/1$ :

1. Parse  $\pi$  as  $(C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha, \pi_Z)$  and compute the Fiat–Shamir challenge  $S_R$  and  $\alpha$ .
2. Verify  $e(C_\phi / C_{S_R}, g_2) = e(\pi_{S_R}, g_2^{V_{S_R}(\tau)})$ .
3. Verify the opening proof  $\pi_\alpha$ :  $e(C_{S_R} / U_\alpha, g_2) = e(\pi_\alpha, g_2^{\tau - \alpha})$ .
4. Compute  $\{L_{i, S'_R}(\alpha)\}_{i \in S'_R}$  and verify the zk-CP-SNARK proof  $\pi_Z$  with public inputs.

**Ver**<sub>key</sub>( $vk, sk$ )  $\rightarrow 0/1$ : For  $sk = s \in \mathbb{F}_p$ , return 1 if and only if  $vk = h^s$  in  $G_1$ .

**Dec**( $F_{[\ell]}, sk, ct$ )  $\rightarrow \{\phi(i)\}_{i \in [\ell]} \cup \{\perp\}$ :

1. Compute the unmasked evaluations  $\{\phi'(i) := ct_i - H(sk, i)\}_{i \in [m]}$ .
2. If  $\text{RS.Det}([m], \{\phi'(i)\}_{i \in [m]}) = 0$  (no error), then output  $\{\phi'(i)\}_{i \in [\ell]}$ .
3. Otherwise, return  $\text{RS.Dec}([\ell], [m], \{\phi'(i)\}_{i \in [m]})$ , including  $\perp$  on failure.

Fig. 1. Practical FDE Scheme

subset [7]. Our construction replaces the prior consistency layer, reusing only this RS-sampling foundation. In  $\text{VECK}_{\text{EL}}^*$  the sampled subset is still carried by



ElGamal ciphertexts—tied to the commitment outside the SNARK, but proven well formed under  $\text{sk}$  inside it. We drop these ciphertexts; instead we

1. derive a sampled set  $S_R$ ,
2. build a low-degree polynomial  $\tilde{\phi}_{S_R}(X)$  agreeing with the sampled evaluations,
3. prove with KZG that  $\tilde{\phi}_{S_R}(X)$  is consistent with the original commitment  $C_\phi$ ,
4. open  $\tilde{\phi}_{S_R}(X)$  at a Fiat–Shamir challenge point  $\alpha$ , exposing only the group-encoded opening  $U_\alpha := g_1^{u_\alpha}$  with  $u_\alpha := \tilde{\phi}_{S_R}(\alpha)$ , and
5. use a lightweight zk-CP-SNARK only to certify that the sampled masked values, the secret key, and  $u_\alpha$  are mutually consistent through the interpolation relation at  $\alpha$ .

Removing the sampled ciphertexts, and with them all curve arithmetic, shrinks the circuit and frees the curve (Section 6), while the group-encoded opening carries the soundness and zero-knowledge arguments below.

The interpolation in step (5) is taken over the augmented domain  $S'_R := S_R \cup \{-1, -2\}$  rather than  $S_R$ : blinding  $\tilde{\phi}_{S_R}$  to hide the sampled evaluations raises its degree to  $|S_R| + 1$ , so fixing it by interpolation requires  $|S_R| + 2$  points, two more than  $S_R$  provides. The auxiliary coordinates  $-1, -2$ —any fixed points outside  $[m]$ —supply these, and the prover publishes the masked values  $ct_{-1}, ct_{-2}$  alongside the file ciphertext.

Figure 1 specifies the full scheme; the next section details its zk-CP-SNARK component  $\pi_Z$ .

## 5.2 The masking relation and CP-linking

The challenge point  $\alpha$  for the KZG opening is also derived by Fiat–Shamir, but unlike an ordinary hash-to-field step it must fall outside  $S'_R = S_R \cup \{-1, -2\}$ . We therefore set

$$\alpha := \text{DerivePoint}_H(C_\phi, \text{vk}, \{ct_i\}_{i \in [m] \cup \{-1, -2\}}, m, R, C_{S_R}, \pi_{S_R}, S_R),$$

where  $\text{DerivePoint}_H$  hashes the transcript with an incrementing counter and returns the first candidate field element not in  $S'_R$ ; since  $R \ll p$  the expected number of trials is  $1 + O(R/p)$ , and  $\alpha \in \mathbb{F}_p \setminus S'_R$ .

The linking gadget  $\text{CPLink}$  enforces two linking equations, binding the in-circuit secret key  $\text{sk}$  to  $\text{vk} = h^{\text{sk}}$  and the in-circuit opening value  $u_\alpha$  to  $U_\alpha = g_1^{u_\alpha}$ ; both  $\text{vk}$  and  $U_\alpha$  are single-base Pedersen commitments, exactly the form the gadget supports. Both equations are linear statements over  $G_1$ , so they are stacked into one Kiltz–Wee instance, so a single proof element discharges both links. Appendix A.1 states the composition proposition. The circuit itself enforces only the hash constraints  $ct_i = x_i + H(\text{sk}, i)$  and the interpolation equation  $u_\alpha = \sum_{i \in S'_R} x_i L_{i, S'_R}(\alpha)$  of the relation  $R_{\text{mask}}$  (Figure 1), while the two group-valued relations are discharged by the linking layer over  $G_1$ , so the circuit performs no elliptic-curve operations. Since the committed  $\text{sk}$  is the same scalar fed to  $H(\text{sk}, i)$ , the link binds the hash input to the key determining  $\text{vk}$ . Thus the CP-SNARK certifies consistency of the sampled masked values, the secret key, and the hidden opening value while revealing neither  $u_\alpha$  nor  $\text{sk}$ .

### 5.3 Design rationale

Two reasons motivate the commit-and-prove choice, specific to the FDE setting.

*Lowering the on-chain cost.* The only on-chain predicate is the key check  $vk = h^{sk}$ , whose gas cost depends on the curve carrying  $vk$ ; Ethereum exposes precompiles only for bn254 and BLS12-381. Proving the key relation *inside* the circuit would force  $vk$  onto the inner curve, outside that set, at higher gas (Section 6.5). Delegating  $vk = h^{sk}$  and  $U_\alpha = g_1^{u_\alpha}$  to CPLink keeps the circuit free of elliptic-curve operations and the public-key group free to be precompile-friendly.

*Enabling the zero-knowledge simulation.* The second reason is internal to the proof: VECK proves correct encryption of *externally* committed data under a universal KZG CRS, so the simulator never learns  $\tau$ , and a *scalar* opening of  $C_{S_R}$  at  $\alpha$  cannot be simulated. The group encoding can: picking  $y_2 \xleftarrow{\$} \mathbb{F}_p$  and setting  $\pi_\alpha = g_1^{y_2}$ ,  $U_\alpha = C_{S_R}/g_1^{y_2(\tau-\alpha)}$  yields a valid opening transcript from the public CRS alone, and CPLink hands the hidden scalar to the CP-SNARK simulator. The commit-and-prove layer thus does double duty: beyond linking, it is what lets the zero-knowledge simulation go through (Appendix A).

*Setup profile and the choice of proving system.* Beyond the external universal KZG CRS shared by all schemes compared here, Gen samples the scheme-specific string  $crs_z$  carrying the CP-SNARK keys—the same profile as  $VECK_{EL}^*$  with its circuit-specific Groth16 setup, whereas the  $\Sigma$ -protocol of Tas et al. is transparent beyond the universal SRS [5]. Nothing in the design commits to Groth16, though. The proof needs two properties of the CP-SNARK from assumption (ii): knowledge soundness, and a simulator that, under a simulated CRS, produces accepting proofs for arbitrary well-formed statements—including false ones. The second is stronger than plain composable zero-knowledge, which simulates only true statements; we need it because intermediate hybrids simulate witness-less statements (Appendix A.7). This requirement is mild, met by every candidate: the CP-Link schemes we build on—Kiltz–Wee [18] and the CRS-updatable QANIZK of Abdolmaleki and Slamanig [23]—simulate arbitrary statements by construction; Groth16 simulates any statement from its CRS trapdoor [21]; and PLONK is even simulation-extractable in the ROM under falsifiable assumptions [24]. Since the circuit uses no elliptic-curve operations, any such commit-and-prove system over the same scalar field can be substituted. We use Groth16 to match the proving stack of  $VECK_{EL}^*$  for a controlled comparison; a universal-setup system like PLONK [22] would remove the circuit-specific ceremony at a modest cost in proof size and verification, with the linking layer re-instantiated for its wire commitments.

### 5.4 Security

Soundness follows from a spot-checking argument: passing the sampled check on enough positions forces agreement with a codeword within the decoding radius, and hence with the committed file. Here the sampled check is realized by KZG subset consistency together with the CP-SNARK. The only extra loss relative

to the ideal sampled check is a Schwartz–Zippel term for the maximum degree  $n$  supported by the KZG CRS.

**Theorem 1 (Security of the construction).** *Work in the algebraic group model (AGM) and the random oracle model (ROM), and let  $q_S, q_P$  bound the adversary’s subset- and point-challenge queries. Assume:*

- (i) *the  $n$ -discrete-logarithm problem for the KZG CRS is hard, so that the KZG scheme is extractable-binding and evaluation-sound for degree at most the CRS-supported bound  $n$ ; we write  $\epsilon_{\text{KZG}}$  for the aggregate advantage of the constant number of reductions to this problem used in Lemma 2;*
- (ii) *(proof-system interface) the commit-and-prove SNARK with its CPLink instance for  $U_\alpha = g_1^{u_\alpha}$  and  $\text{vk} = h^{\text{sk}}$  (link key on bases  $(g_1, h)$ ) is complete, knowledge-sound (error  $\epsilon_{\text{CP}}^{\text{snd}}$ ), and composable zero-knowledge with arbitrary-statement simulation (error  $\epsilon_{\text{CP}}^{\text{zk}}$ );*
- (iii) *(algebraic CRS compatibility) the auxiliary CP-SNARK setup is  $\tau$ -algebraically compatible with the KZG CRS (Definition 5; realized by Corollaries 1–2);*
- (iv) *the discrete-logarithm problem is hard in  $G_1$  with advantage at most  $\epsilon_{\text{DL}}^{G_1}$ ;*
- (v)  *$H$  is modeled as a random oracle; and*
- (vi) *the Reed–Solomon detector RS.Det and decoder RS.Dec are correct within the decoding radius  $t = \lfloor (m - \ell)/2 \rfloor$ .*

*If the redundancy factor satisfies  $q_S((\beta + 1)/(2\beta))^R \leq 2^{-\lambda}$ , then the construction of Figure 1 is a secure VECK in the sense of Tas et al. [5]: it is correct, sound, and computationally zero-knowledge. Quantitatively, every probabilistic polynomial-time algebraic adversary respecting these grinding budgets has soundness advantage—the probability of producing an accepting transcript whose revealed key decrypts to a file different from the committed one—at most*

$$\epsilon_{\text{KZG}} + \epsilon_{\text{CP}}^{\text{snd}} + q_P \frac{n}{p - R - 2} + q_S \left( \frac{\beta + 1}{2\beta} \right)^R,$$

*where the two factors separately account for Fiat–Shamir grinding over  $S_R$  and  $\alpha$ , and zero-knowledge distinguishing advantage at most*

$$\epsilon_{\text{CP}}^{\text{zk}} + \epsilon_{\text{DL}}^{G_1} + \frac{q_P}{p}.$$

Here  $C_\phi$  is the honestly generated degree- $\ell$  commitment of the model and  $F(\phi) = (\phi(0), \dots, \phi(\ell))$  is full-file retrieval.

*Proof (Proof overview).* The three VECK properties are proved in full in Appendix A (Lemmas 3–5) and assembled in Appendix A.8.

*Correctness* holds by inspection: the blinded  $\tilde{\phi}_{S_R}$  agrees with  $\phi$  on  $S_R$  with degree at most  $|S_R| + 1$ , so  $S'_R$  supplies the  $|S_R| + 2$  interpolation points, the two KZG checks and the CP-SNARK relation verify, and honest unmasking yields the exact codeword.

*Soundness.* CP-SNARK knowledge soundness extracts  $\text{sk}, u_\alpha, \{x_i\}_{i \in S'_R}$  satisfying the masking relation and the two CPLink equations; here  $U_\alpha = g_1^{u_\alpha}$  turns

the group element into the *explicit* scalar opening value to which KZG evaluation soundness applies (Lemma 2), and  $\mathbf{vk} = h^{\mathbf{sk}}$  pins the extracted key to the  $\log_h \mathbf{vk}$  used by the decryptor. A Schwartz–Zippel argument at  $\alpha$  then forces  $ct_i - H(\mathbf{sk}, i) = \phi(i)$  on  $S_R$ , and the Reed–Solomon sampling bound catches any accepted word beyond the decoding radius except with probability  $((\beta+1)/2\beta)^R$  per sample; union-bounding over  $q_S$  subset and  $q_P$  point challenges gives the stated bound.

*Zero-knowledge* is a five-hybrid argument (Lemma 5). The simulator, which never learns  $\tau$ , replaces  $\pi_Z$  by a simulated proof, the hash-masked word by uniform field elements (the only bad event, a prior query  $H(\mathbf{sk}, \cdot)$ , reduces to discrete log in  $G_1$ ), and the KZG opening and subset transcripts by uniform valid pairing transcripts from the public CRS alone—using the group-element opening  $U_\alpha$ , not the scalar  $u_\alpha$ .

*Assumption footprint.* The models match the prior VECK constructions [5, 7], likewise proven secure in the AGM and ROM, but the computational assumptions are strictly fewer: the ciphertext is a random-oracle mask rather than a public-key encryption, so the DDH assumption of the ElGamal layers [5, 7] and the DCR assumption of the Paillier variant [5] are dropped. Every computational assumption is of discrete-logarithm type:  $n$ -DLOG for the KZG layer (Lemma 2), and—under our instantiation—DL for the masking and Kiltz–Wee links and the  $q$ -DLOG-type assumption behind Groth16’s AGM analysis [18, 21].

## 6 Evaluation

### 6.1 Experimental Setup

We implemented KZG in Rust/arkworks and the SNARK in Go/gnark, following the stack of  $\text{VECK}_{\text{EL}}^*$  [7].<sup>4</sup> We implemented the linking layer for  $U_\alpha = g_1^{u_\alpha}$  and  $\mathbf{vk} = h^{\mathbf{sk}}$  as a single stacked Kiltz–Wee instance [18]. Experiments ran on an **Apple M3 Pro (18 GB** memory, up to **11 cores**) with **Groth16** [21] and **Poseidon2** in the circuit [25]. We benchmark on two curves: BLS12-381, which our construction is free to use because it places no elliptic-curve gadgets in the circuit, and BW6-761, the curve  $\text{VECK}_{\text{EL}}^*$  is forced onto; reporting both isolates the effect of the curve choice from that of the constraint reduction. We fix  $\lambda = 128$ ,  $q_S, q_P \leq 2^{32}$ , and  $R \in \{256, 512, 1024\}$ . The grinding-aware condition  $q_S((\beta+1)/2\beta)^R \leq 2^{-128}$  gives  $\beta_{\min} = 3.37, 1.64, 1.26$ , respectively. For another budget  $q_S = 2^g$ , the formula in Section 4 replaces 32 by  $g$ ; for example  $g = 0$  gives 2.41, 1.47, 1.20. One attempt means hashing a complete candidate transcript.

### 6.2 SNARK Microbenchmark

Table 1 reports the SNARK component’s resource profile—R1CS constraint count, Groth16 setup time, online proving time, verification time, and CRS size on both curves.

<sup>4</sup> Our implementation is available at <https://github.com/dongwook1214/practical-fde>.

**Table 1.** SNARK resource costs for  $\text{VECK}_{\text{EL}}^*$  and our scheme

| scheme (curve)                        | metric            | $R = 256$ | $R = 512$ | $R = 1024$ |
|---------------------------------------|-------------------|-----------|-----------|------------|
| $\text{VECK}_{\text{EL}}^*$ (BW6-761) | constraints       | 1,029,847 | 2,057,687 | 4,113,367  |
|                                       | setup time        | 188.8 s   | 384.3 s   | 772.6 s    |
|                                       | proving time      | 10.87 s   | 20.14 s   | 40.20 s    |
|                                       | verification time | 14.32 ms  | 23.40 ms  | 30.92 ms   |
|                                       | CRS size          | 402.2 MiB | 803.9 MiB | 1.57 GiB   |
| Ours (BW6-761)                        | constraints       | 52,525    | 103,726   | 206,128    |
|                                       | setup time        | 11.51 s   | 22.88 s   | 45.71 s    |
|                                       | proving time      | 1.148 s   | 2.261 s   | 3.973 s    |
|                                       | verification time | 9.77 ms   | 16.31 ms  | 21.59 ms   |
|                                       | CRS size          | 23.90 MiB | 47.34 MiB | 94.23 MiB  |
| Ours (BLS12-381)                      | constraints       | 52,223    | 103,424   | 205,826    |
|                                       | setup time        | 3.40 s    | 6.67 s    | 13.09 s    |
|                                       | proving time      | 321.0 ms  | 595.3 ms  | 1.115 s    |
|                                       | verification time | 2.44 ms   | 3.84 ms   | 5.88 ms    |
|                                       | CRS size          | 14.35 MiB | 28.48 MiB | 56.73 MiB  |

*Constraint count.* Our circuit reduces the number of constraints by roughly  $20\times$  (e.g., 52,223 versus 1,029,847 at  $R = 256$ ), consistent with the main design goal of eliminating asymmetric cryptographic operations from the proving circuit.

*Two sources of the proving speedup.* The BW6-761 rows separate the two effects behind the prover-side improvement. On the *same* BW6-761 curve as  $\text{VECK}_{\text{EL}}^*$ , our scheme already proves 9–10 $\times$  faster (1.148 s vs. 10.87 s at  $R = 256$ )—the pure effect of the constraint reduction. Moving to BLS12-381, which  $\text{VECK}_{\text{EL}}^*$  cannot use because its in-circuit ElGamal ciphertexts must share the KZG group and thus force a pairing-friendly inner curve, yields a further  $\approx 3.6\times$  (1.148 s  $\rightarrow$  321 ms). The two factors compound to more than  $30\times$  for this component (e.g., 40.20 s  $\rightarrow$  1.115 s at  $R = 1024$ ), despite our BLS12-381 scalar field being about two bits larger than the BLS12-377 field  $\text{VECK}_{\text{EL}}^*$  uses.

*Setup time and CRS size.* The constraint reduction carries into setup and key material. At  $R = 256$  on BLS12-381, Groth16 setup drops from 188.8 s to 3.40 s ( $\approx 55\times$ ) and the CRS from 402.2 to 14.35 MiB ( $\approx 28\times$ ), with both gaps persisting across the range (1.57 GiB vs. 56.73 MiB at  $R = 1024$ ); even on BW6-761 the cuts exceed an order of magnitude, confirming they stem from the circuit, not the curve. These are the scheme-specific proof CRS; all schemes additionally use the shared universal KZG CRS, occupying  $48(n+1) + 96(R+1)$  bytes (about 192.1 MiB for  $n = 2^{22}$ ,  $R = 1024$ ) with compressed BLS12-381 elements.

### 6.3 Verification Time

Table 2 compares verification time after the ciphertext transcript and sampled entries are fixed.  $\text{VECK}_{\text{EL}}^+$  must check a  $\Theta(\lambda)$ -sized collection of range proofs,

**Table 2.** Core cryptographic verification time (ms) for  $\text{VECK}_{\text{EL}}^+$ ,  $\text{VECK}_{\text{EL}}^*$ , and our scheme

| scheme (curve)                          | $R = 256$ | $R = 512$ | $R = 1024$ |
|-----------------------------------------|-----------|-----------|------------|
| $\text{VECK}_{\text{EL}}^+$ (BLS12-381) | 1171      | 2353      | 4680       |
| $\text{VECK}_{\text{EL}}^*$ (BW6-761)   | 38        | 47        | 54         |
| Ours (BW6-761)                          | 28        | 38        | 51         |
| Ours (BLS12-381)                        | 9         | 12        | 20         |

since its exponential ElGamal requires proving that each sampled plaintext lies in the brute-forceable decryption range—a substantial cost.  $\text{VECK}_{\text{EL}}^*$  avoids the range proofs (its sampled ElGamal ciphertexts are not the scheme’s own), and our scheme removes ElGamal entirely.

Our scheme is nonetheless the fastest to verify, and the two BW6-761 rows again separate the two factors: on the same curve our scheme already verifies faster (28 vs. 38 ms at  $R = 256$ ), since it checks fewer proof objects, and moving to the unconstrained BLS12-381 brings verification to 9–20 ms across the range.

*CP-linking overhead.* All times reported above already include the commit-and-prove linking layer for  $U_\alpha = g_1^{u_\alpha}$  and  $\text{vk} = h^{\text{sk}}$ , instantiated following Kiltz–Wee [18], which contributes only a small fraction of the total. Generating the linking proof takes under 1 ms, and its verification accounts for 3.4 ms on BW6-761 and 1.2 ms on BLS12-381.

*Proof size and communication overhead.* The proof contains four  $G_1$  elements ( $C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha$ ) plus  $\pi_Z$ , which under our Groth16 instantiation adds four  $G_1$  and one  $G_2$ , so  $\pi = 8G_1 + G_2 = 480$  compressed bytes. The exchange also sends the  $m + 3$  field elements of  $ct$  (including positions  $-1, -2$ ), so  $ct + \pi \approx 32(m + 3) + 480$  bytes, plus 48 if  $\text{vk}$  is not already posted; the  $O(R)$  verifier inputs are derived, not resent. The communication factors are 3.37, 1.64, 1.26 at  $R = 256, 512, 1024$ .

#### 6.4 Online Proof Generation Time

Figure 2 reports the cryptographic proof path—the KZG commitment, the SNARK, and the CP-linking layer—which is the only part our construction changes. The remaining stages (Reed–Solomon extension, masking) are shared and *identical* to  $\text{VECK}_{\text{EL}}^+$ . For  $\text{VECK}_{\text{EL}}^+$  we likewise plot proof generation alone and exclude its encryption cost; this exclusion is generous to  $\text{VECK}_{\text{EL}}^+$ , since it encrypts with full-file exponential ElGamal, which in practice makes its encryption more than  $1000\times$  slower and its communication more than  $10\times$  larger than ours.

At  $2^{17}$  elements, proof generation drops from 21.1 to 0.945 s at  $R = 512$  and from 41.5 to 1.69 s at  $R = 1024$ ; on BW6-761 the saving is about 18 s and 37 s across  $2^{10}$ – $2^{22}$  elements.

The advantage compounds with  $R$ . Because our proof path is so much cheaper, raising  $R$  costs us little, whereas for  $\text{VECK}_{\text{EL}}^*$  it is prohibitive. A larger  $R$  in

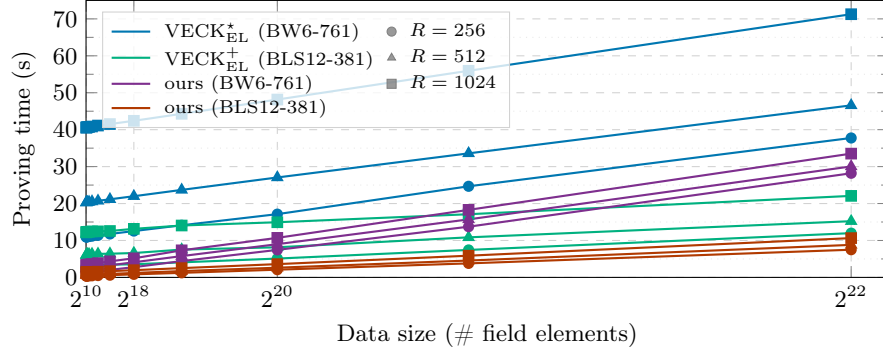


Fig. 2. Proof-generation time on BLS12-381 and BW6-761

Table 3. On-chain gas cost of one  $\text{Ver}_{key}$  scalar multiplication on each candidate curve

| curve     | EVM precompile | gas       | used by                                   |
|-----------|----------------|-----------|-------------------------------------------|
| bn254     | yes            | 6,000     | [5], $\text{VECK}_{\text{EL}}^+$ , ours   |
| BLS12-381 | yes            | 12,000    |                                           |
| BLS12-377 | no             | 920,547   | $\text{VECK}_{\text{EL}}^*$ (BW6-761)     |
| MNT4-753  | no             | 5,894,791 | $\text{VECK}_{\text{EL}}^*$ (MNT 2-chain) |

turn admits a more aggressive redundancy  $\beta$ —recall  $\beta_{\min} = 3.37, 1.64, 1.26$  at  $R = 256, 512, 1024$ —and since the communication factor equals  $\beta$ , our scheme can operate in the large- $R$ , low- $\beta$  regime that keeps proving fast and communication small at once.  $\text{VECK}_{\text{EL}}^*$ , pinned to small  $R$  by its proving cost, is left paying the higher  $\beta$  and hence larger communication.

### 6.5 EVM Cost of the Key-Verification Group Operation

$\text{Ver}_{key}$ ’s group operation is one scalar multiplication, on precompiled bn254 or BLS12-381 in the original VECK [5],  $\text{VECK}_{\text{EL}}^+$ , and ours, but on non-precompiled BLS12-377 or MNT4-753 for  $\text{VECK}_{\text{EL}}^*$ . On the non-precompiled curves we implement the scalar multiplication in the EVM with mixed Jacobian–affine addition and a width-4 windowed method, so the reported gas figures reflect an optimized rather than naive implementation. By Table 3, BLS12-377 costs 920,547 gas and MNT4-753 5,894,791, versus 12,000 on BLS12-381; counts exclude call-data, memory, validation, equality, and contract logic.

## 7 Discussion and Limitations

Removing in-circuit public-key operations cuts the sample-size SNARK term, leaving KZG as the next bottleneck. The construction retains a universal SRS, a grinding-aware redundancy trade-off, and a heuristic ROM Poseidon2. Benchmarks isolate proof costs, leaving encoding, RS work, network, peak memory.

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## A Complete Security Proofs

This appendix proves Theorem 1 in full: after fixing the security model (Appendix A.1) we prove an algebraic-group extraction lemma for the KZG layer (Appendix A.3), then correctness, soundness, and zero-knowledge (Appendices A.4–A.7), assembled in Appendix A.8.

We work throughout in the algebraic group model (AGM) and the random oracle model (ROM). In the AGM every group element an adversary outputs comes with a representation as a known exponent-linear combination of the elements it received; Lemma 2 uses these representations directly for the KZG layer, while the CP-SNARK and CPLink witnesses come from the knowledge extractors assumed in Theorem 1(ii), which our Groth16 and Kiltz–Wee instantiations provide in this model. We write  $\tau$  for the (secret, external) KZG trapdoor, fix  $h := g_1$ , and write  $n$  for the CRS-supported degree bound and  $p = |\mathbb{F}_p|$ ; recall  $S'_R = S_R \cup \{-1, -2\}$  with  $|S_R| = R$  and  $|S'_R| = R + 2$ . The standing constraints  $n \geq \max(\ell, R + 1)$  and  $\lceil \beta(\ell + 1) \rceil \geq R$ —ensuring the CRS supports both  $\phi$  and the degree- $(R + 1)$  blinded  $\tilde{\phi}_{S_R}$ , and that an  $R$ -subset of  $[m]$  exists—hold in the regime  $R \ll \ell \leq m \leq n$ .

### A.1 Security model and assumptions

In the FDE instantiation the committed object is the file polynomial  $w = \phi \in \mathbb{F}_p^{\leq \ell}[X]$ , its commitment is the honestly generated KZG commitment  $C_\phi = \text{Commit}(crs, \phi) = g_1^{\phi(\tau)}$ , and the function to be encrypted is  $F(\phi) = (\phi(0), \dots, \phi(\ell))$ . We use the VECK security notions of Tas et al. [5], which we restate here specialized to this  $F$  so that the appendix is self-contained.

**Definition 1 (Correctness).** *For every  $\phi$  of degree at most  $\ell$ ,*

$$\Pr \left[ \begin{array}{l} \text{Ver}_{ct}(F, C_\phi, \text{vk}, ct, \pi) = 1 \wedge \\ \text{Ver}_{key}(\text{vk}, \text{sk}) = 1 \end{array} \middle| \begin{array}{l} crs \leftarrow \text{Setup}(1^\lambda); C_\phi \leftarrow \text{Commit}(crs, \phi) \\ pp \leftarrow \text{Gen}(crs) \\ (\text{vk}, \text{sk}, ct, \pi) \leftarrow \text{Enc}(F, C_\phi, \phi) \end{array} \right] = 1,$$

*and  $\text{Dec}(F, \text{sk}, ct) = F(\phi)$  with probability 1.*

**Definition 2 (Soundness).** *A scheme is sound with advantage  $\epsilon_{\text{snd}}$  if for every  $\phi$  of degree at most  $\ell$  and every PPT (algebraic) adversary  $\mathcal{A}$ ,*

$$\Pr \left[ \begin{array}{l} \text{Ver}_{ct}(F, C_\phi, \text{vk}, ct, \pi) = 1 \wedge \\ \text{Ver}_{key}(\text{vk}, \text{sk}) = 1 \wedge y \neq F(\phi) \end{array} \middle| \begin{array}{l} crs \leftarrow \text{Setup}(1^\lambda); pp \leftarrow \text{Gen}(crs) \\ C_\phi \leftarrow \text{Commit}(crs, \phi) \\ (\text{sk}, \text{vk}, ct, \pi) \leftarrow \mathcal{A}^H(pp, F, \phi, C_\phi) \\ y \leftarrow \text{Dec}(F, \text{sk}, ct) \end{array} \right] \leq \epsilon_{\text{snd}}.$$

**Definition 3 (Computational zero-knowledge).** *A scheme is computationally zero-knowledge with advantage  $\epsilon_{\text{zk}}$  if there is a PPT simulator  $\text{Sim}$  (with*

oracle control of  $H$ ) such that for every  $\phi$  of degree at most  $\ell$  and every PPT distinguisher  $\mathcal{D}$ ,

$$\left| \Pr[\mathcal{D}^H(pp, C_\phi, \text{vk}, ct, \pi) = 1] - \Pr[\mathcal{D}^H(\widetilde{pp}, C_\phi, \widetilde{\text{vk}}, \widetilde{ct}, \widetilde{\pi}) = 1] \right| \leq \epsilon_{\text{zk}},$$

where the real experiment samples  $pp = (crs, crs_z, h)$  by  $\text{Gen}$  and runs  $\text{Enc}$ . In the simulated experiment the external KZG  $crs$  and  $h = g_1$  are identical, while  $(\widetilde{crs}_z, \text{st}) \leftarrow S_{\text{kg}}^{\text{CP}}(g_1, h, R)$ ,  $\widetilde{pp} := (crs, \widetilde{crs}_z, h)$ , and  $(\widetilde{\text{vk}}, \widetilde{ct}, \widetilde{\pi}) \leftarrow \text{Sim}(\widetilde{pp}, \text{st}, F, C_\phi)$ . The simulator receives public  $F$  but neither  $\phi$  nor  $\tau$ , and the view is taken before key revelation. Indistinguishability of the honest and simulated CP reference strings is included in the definition. This formulation is equivalent, up to the standard factor 2, to the bit-guessing form of Tas et al. [5].

**Definition 4 ( $n$ -DLOG).** For  $\tau \xleftarrow{\$} \mathbb{F}_p$ , given the challenge  $(\mathcal{G}, \{g_1^{\tau^i}\}_{i=0}^n, \{g_2^{\tau^i}\}_{i=0}^R)$ , output  $\tau$ . An algorithm's advantage  $\epsilon_{n\text{-dl}}$  is its probability of success. The powers exposed match the KZG CRS of Lemma 2 exactly—degree  $n$  in  $G_1$  and degree  $R$  in  $G_2$ —and this is the standard  $q$ -type discrete-logarithm assumption that underlies KZG extractable binding and evaluation soundness in the AGM [10]. It is a falsifiable, computational assumption: there is no averaging over  $\tau$ , which is fixed by the challenge.

We isolate the property of the auxiliary (CP-SNARK) setup that Lemma 2 actually needs, keeping it separate from the proof-system interface of Theorem 1(ii).

**Definition 5 ( $\tau$ -algebraically compatible auxiliary CRS).** An auxiliary proof-system setup producing  $G_1$ -elements  $crs_z = (Z_1, \dots, Z_k)$  is  $\tau$ -algebraically compatible with a KZG CRS  $(\{g_1^{\tau^i}\}_{i=0}^n, \{g_2^{\tau^i}\}_{i=0}^R)$  if there is a PPT setup simulator (equivalently, a representation algorithm) that outputs, for every  $Z_j$ , a polynomial  $f_j \in \mathbb{F}_p[X]$  of degree at most  $d_{\text{aux}} \leq n$  with

$$Z_j = g_1^{f_j(\tau)},$$

such that the resulting joint public-parameter distribution  $(crs, crs_z)$  is identical to, or computationally indistinguishable from, the real setup distribution. The  $f_j$  are known to the reduction;  $\tau$  is not.

This one definition subsumes the independent, shared-SRS, and transparent setups, each realized by a distinct representation algorithm.

**Corollary 1 (Independent setup).** Independently generated Groth16 and Kiltz–Wee parameters are  $\tau$ -algebraically compatible with  $d_{\text{aux}} = 0$ , provided the reduction generates the auxiliary setup itself and retains the exponent representation of each  $G_1$ -element: every  $Z_j = g_1^{\zeta_j}$  for a known  $\tau$ -free scalar  $\zeta_j$ , i.e.  $f_j \equiv \zeta_j$ . A transparent setup falls in this case, the reduction programming the setup's random oracle (domain-separated from  $H$ ) to place each  $Z_j = g_1^{\zeta_j}$ , which preserves the distribution.

**Corollary 2 (Shared SRS).** *A shared-SRS proving system whose commitment key reuses the KZG powers is  $\tau$ -algebraically compatible whenever every auxiliary  $G_1$ -element is accompanied by an efficiently computable representation in the KZG power basis  $\{\tau^i\}_{i=0}^n$ ; then  $d_{\text{aux}} \leq n$ , and the reduction reuses the challenge CRS verbatim without knowing  $\tau$ .*

*Proof.* In each case the stated representation algorithm returns polynomials  $f_j$  of degree at most  $d_{\text{aux}} \leq n$  with  $Z_j = g_1^{f_j(\tau)}$ . Corollary 1: the reduction samples the auxiliary trapdoor, so each  $\zeta_j$  is known and  $\tau$ -free, and self-generation reproduces the real distribution exactly. Corollary 2: the supplied basis representation gives  $f_j$  directly, and reusing the challenge powers leaves the distribution unchanged. In both the joint distribution matches the real one, meeting Definition 5.

*Assumptions.* We use the assumptions of Theorem 1:  $n$ -DLOG (Definition 4) in the bilinear group (advantage  $\epsilon_{n\text{-dl}}$ ), which underlies KZG extractable binding and evaluation soundness in the AGM, and ordinary discrete-logarithm hardness in  $G_1$  (advantage  $\epsilon_{\text{DL}}^1$ ); the proof-system interface—knowledge soundness, equality linking, arbitrary-statement simulation under the simulated CRS, and composable zero-knowledge of the CP-SNARK and CPLink (errors  $\epsilon_{\text{CP}}^{\text{snd}}, \epsilon_{\text{CP}}^{\text{zk}}$ ; for the link, see [18, 15]);  $\tau$ -algebraic compatibility of the auxiliary CP CRS with the KZG CRS (Definition 5), which is exactly the representation property Lemma 2 requires and which the settings of Corollaries 1–2 realize; and the ROM for  $H$ . Since  $h = g_1$ , Lemma 2 reduces only to  $n$ -DLOG, so  $\epsilon_{\text{KZG}} \leq c\epsilon_{n\text{-dl}}$  for a small constant  $c$ . The KZG CRS is external, with  $\tau$  unknown to all reductions and to the simulator.

## A.2 The proof-system interface: arbitrary-statement simulation

Assumption (ii) of Theorem 1 asks two things of the commit-and-prove layer: knowledge soundness and *arbitrary-statement simulation*. The second is stronger than the composable zero knowledge that NIZK definitions usually state, and the zero-knowledge argument (Appendix A.7) needs it: hybrids  $H_2$ – $H_4$  simulate proofs for statements that carry no witness. We first pin the property down as a game, then show it composes from the link and the core proof, and finally exhibit two named instantiations—one trusted, one universal—each of which satisfies the whole of assumption (ii)–(iii).

**Definition 6 (Arbitrary-statement simulation).** *Let  $\Pi = (\text{KeyGen}, \text{Prove}, \text{VerProof})$  prove a relation  $\mathcal{R}$  with a well-formedness predicate  $\text{WF}$ , where  $\text{WF}(x) = 1$  means  $x$  lies in the domain of  $\text{VerProof}$  (commitments in the right groups, public inputs in  $\mathbb{F}_p$ ), regardless of whether  $x$  is true.  $\Pi$  has arbitrary-statement simulation with error  $\epsilon^{\text{zk}}$  if there is a simulator pair  $(S_{\text{kg}}, S_{\text{prv}})$  such that:*

- (a) (simulated setup)  $(\widetilde{\text{crs}}, \text{td}) \leftarrow S_{\text{kg}}(pp)$  yields a reference string  $\widetilde{\text{crs}}$  perfectly, statistically, or computationally indistinguishable from an honest  $\text{crs} \leftarrow \text{KeyGen}(pp)$ ;

- (b) (simulation completeness on all well-formed statements) for every  $x$  with  $\text{WF}(x) = 1$ —in particular for false  $x$ —the proof  $\text{S}_{\text{prv}}(\widetilde{\text{crs}}, \text{td}, x)$  is accepted by  $\text{VerProof}(\widetilde{\text{crs}}, x, \cdot)$  with probability 1, using no witness; and
- (c) (indistinguishability on true statements) for every  $(x, w) \in \mathcal{R}$ ,  $(\widetilde{\text{crs}}, \text{S}_{\text{prv}}(\widetilde{\text{crs}}, \text{td}, x))$  and  $(\text{crs}, \text{Prove}(\text{crs}, x, w))$  are within  $\epsilon^{\text{zk}}$ .

Composable zero knowledge is the special case in which clause (b) is demanded only for true  $x$ ; clause (b) for false  $x$  is the sole extra requirement, and it is purely syntactic—an accepting proof exists for every statement in the verifier’s domain. Indistinguishability (clause (c)) is never asked on a false statement.

The next lemma is the modular core of the interface: it builds the property from the two ingredients we actually implement, isolating exactly where the committed witness enters.

**Lemma 1 (Composition preserves the interface).** *Realize the commit-and-prove layer from*

- a core proof system  $\text{CP}_{\text{core}}$  proving  $\mathcal{R}_{\text{mask}}$  over an internal witness commitment  $cc$  whose first coordinates are  $(u_\alpha, \text{sk})$ ; and
- a link  $\text{CP}_{\text{link}}$  selecting those coordinates of  $cc$  and binding them to the external commitments  $U_\alpha = g_1^{u_\alpha}$  and  $\text{vk} = h^{\text{sk}}$  on the bases  $(g_1, h)$ ,

sharing the computationally hiding commitment  $cc$ . If  $\text{CP}_{\text{core}}$  and  $\text{CP}_{\text{link}}$  are each complete, knowledge-sound (errors  $\epsilon_{\text{core}}^{\text{snd}}, \epsilon_{\text{link}}^{\text{snd}}$ ), and have arbitrary-statement simulation (Definition 6, errors  $\epsilon_{\text{core}}^{\text{zk}}, \epsilon_{\text{link}}^{\text{zk}}$ ), then their composition satisfies Theorem 1(ii) with

$$\epsilon_{\text{CP}}^{\text{snd}} \leq \epsilon_{\text{core}}^{\text{snd}} + \epsilon_{\text{link}}^{\text{snd}}, \quad \epsilon_{\text{CP}}^{\text{zk}} \leq \epsilon_{\text{core}}^{\text{zk}} + \epsilon_{\text{link}}^{\text{zk}} + \epsilon_{\text{hide}},$$

where  $\epsilon_{\text{hide}}$  is the hiding advantage of  $cc$ .

*Proof. Knowledge soundness.* On an accepting composed proof  $(cc, \pi_{\text{core}}, \pi_{\text{link}})$  the core extractor returns  $((u_j), \omega)$  with  $cc = \text{Commit}((u_j); \rho)$  and  $\mathcal{R}_{\text{mask}}((u_j), \omega) = 1$ ; the link extractor returns openings of the *same*  $cc$  and of  $U_\alpha, \text{vk}$  with equal selected coordinates. Binding of  $cc$  forces the two openings of its selected coordinates to agree, so  $U_\alpha = g_1^{u_\alpha}$  and  $\text{vk} = h^{\text{sk}}$  for the extracted  $(u_\alpha, \text{sk})$ ; a union bound gives the stated error.

*Arbitrary-statement simulation.* Set  $\text{S}_{\text{kg}}^{\text{CP}}$  to run  $\text{S}_{\text{kg}}^{\text{core}}$  and  $\text{S}_{\text{kg}}^{\text{link}}$  and return the pair of simulated strings  $\widetilde{\text{crs}}$  with their simulation states  $\text{td}$ ; the joint string is indistinguishable from the honest joint string by a two-step hybrid over clause (a) of the two components. For a well-formed composed statement  $x = (\text{public inputs}, U_\alpha, \text{vk})$ ,  $\text{S}_{\text{prv}}^{\text{CP}}$  samples a well-formed commitment  $cc \leftarrow \text{Commit}(\mathbf{0}; \rho)$ , forms the sub-statements  $x_{\text{core}} = (\text{public inputs}, cc)$  and  $x_{\text{link}} = (cc, U_\alpha, \text{vk})$ , and outputs  $(cc, \text{S}_{\text{prv}}^{\text{core}}(\cdot, x_{\text{core}}), \text{S}_{\text{prv}}^{\text{link}}(\cdot, x_{\text{link}}))$ . Both sub-statements are well-formed whenever  $x$  is, so Definition 6(b) for each component makes both sub-proofs accept regardless of truth; the composed proof therefore verifies for every well-formed  $x$ , false ones included. This is clause (b) for the composition.

For clause (c), fix a true  $(x, w)$  and walk the hybrids:  $H_0$  is the honest proof with  $cc^* = \text{Commit}((u_j); \rho)$ ;  $H_1$  simulates the core setup and proof—the core

statement is true, so Definition 6(c) for the core costs  $\epsilon_{\text{core}}^{\text{zk}}$ ;  $H_2$  simulates the link setup and proof—likewise  $\epsilon_{\text{link}}^{\text{zk}}$ ;  $H_3$  replaces  $cc^*$  by  $\text{Commit}(\mathbf{0}; \rho)$ . In  $H_2$  both sub-proofs are produced from the simulation states  $\text{td}$  and statements alone, so the opening of  $cc$  appears nowhere in the view;  $H_2$  and  $H_3$  differ only in the committed value, bounded by  $\epsilon_{\text{hide}}$ .  $H_3$  is the composed simulator. Summing the three steps gives the bound; completeness is immediate.

The proof locates each part of the interface: clause (b) is the syntactic guarantee that an accepting proof exists for any statement the verifier accepts as well-formed; clause (c) confines indistinguishability to true statements; and the committed witness enters *only* through the final hiding step. Two named instantiations discharge the interface together with the compatibility of assumption (iii).

**Corollary 3 (Trusted-setup instantiation: Groth16 + Kiltz–Wee).** *Instantiate  $\text{CP}_{\text{core}}$  with Groth16 in its committed-witness (LegoSNARK) form [21, 15] and  $\text{CP}_{\text{link}}$  with the Kiltz–Wee QA-NIZK for linear subspaces [18] on the bases  $(g_1, h)$ . Then, in the AGM:*

- (i) Knowledge soundness. *Groth16 is knowledge-sound under its  $q$ -type ( $q$ -DLOG) assumption; the Kiltz–Wee link is knowledge-sound for witness-samplable subspace matrices under discrete log, which our two single-base equality relations meet. Lemma 1 lifts these to the composition.*
- (ii) Arbitrary-statement simulation. *The Groth16 simulator produces an accepting proof for any statement from the CRS trapdoor (perfect); the Kiltz–Wee simulator does likewise from the QA-NIZK trapdoor (perfect zero knowledge). The LegoSNARK witness commitment is perfectly hiding ( $\epsilon_{\text{hide}} = 0$ ). By Lemma 1 the composition has arbitrary-statement simulation.*
- (iii) Algebraic CRS compatibility. *Both strings are sampled independently of  $\tau$ ; the reduction generates them and keeps each exponent, so  $f_j \equiv \zeta_j$  with  $d_{\text{aux}} = 0$  (Corollary 1).*

*Hence this instantiation—the one benchmarked in Section 6—meets Theorem 1(ii)–(iii); the setup is trusted and circuit-specific. Its core proof is  $2G_1 + G_2$ , and the committed witness element and one stacked Kiltz–Wee link add  $2G_1$ , so  $\pi_Z = 4G_1 + G_2$ .*

The interface is not tied to a trusted setup. A universal instantiation—PLONK over the external KZG SRS [22] for the core, and the CRS-updatable asymmetric QA-NIZK of Abdolmaleki and Slamanig [23] for the link—appears to satisfy the same three requirements, and would remove the circuit-specific ceremony. PLONK is knowledge-sound (indeed simulation-extractable in the ROM under a falsifiable  $q$ -DLOG-type assumption [24]) and, by the same work, admits a *trapdoorless* simulator that handles arbitrary statements in the ROM, while the QA-NIZK is knowledge-sound under BDH-KE—which is what makes it LegoSNARK-composable—and its simulator produces an accepting proof for every statement from its trapdoor. For  $\tau$ -compatibility, PLONK reuses the KZG powers (Corollary 2), and the QA-NIZK string is algebraically independent of  $\tau$ —the  $\tau$ -free case of Corollary 1. We do not carry out this instantiation in full:

under an updatable ceremony the reduction no longer samples the QA-NIZK string itself, so its  $\tau$ -free representation must instead be recovered by extracting the update contributions—e.g. through BDH-KE—and this has to be checked against the concrete update procedure. We leave a complete universal instantiation to future work.

### A.3 Algebraic extraction for the KZG layer

The next lemma is the rigorous form of the two KZG checks in  $\text{Ver}_{ct}$ . Part (a) is the subset-consistency guarantee; part (b) is evaluation soundness, stated so that it binds the scalar  $u_\alpha$  once that scalar is made explicit (which, in the construction, is the role of the CPLink equation  $U_\alpha = g_1^{u_\alpha}$ ).

**Lemma 2 (KZG soundness in the AGM).** *Let  $crs = (\mathcal{G}, \{g_1^{\tau^i}\}_{i=0}^n, \{g_2^{\tau^i}\}_{i=0}^R)$  with  $\tau \xleftarrow{\$} \mathbb{F}_p$ ,  $h := g_1$ , and  $C_\phi = g_1^{\phi(\tau)}$  for a known  $\phi$  of degree at most  $\ell \leq n$ . Let  $\mathcal{A}$  be an algebraic algorithm that outputs  $C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha \in G_1$  (with representations), a set  $S_R \subseteq [m]$  of size  $R \leq n$ , and a point  $\alpha \in \mathbb{F}_p \setminus S'_R$ , such that the two checks*

$$e(C_\phi/C_{S_R}, g_2) = e(\pi_{S_R}, g_2^{V_{S_R}(\tau)}), \quad e(C_{S_R}/U_\alpha, g_2) = e(\pi_\alpha, g_2^{\tau-\alpha})$$

*hold. Then there is a polynomial  $\hat{\phi}_{S_R} \in \mathbb{F}_p[X]$  of degree at most  $n$  such that, except with probability  $\epsilon_{\text{KZG}} \leq c \epsilon_{n\text{-dl}}$  for a small constant  $c$ ,*

- (a)  $C_{S_R} = g_1^{\hat{\phi}_{S_R}(\tau)}$  and  $\hat{\phi}_{S_R}(i) = \phi(i)$  for all  $i \in S_R$ ; and
- (b) for any scalar  $y \in \mathbb{F}_p$  known to the reduction with  $U_\alpha = g_1^y$ , we have  $\hat{\phi}_{S_R}(\alpha) = y$ .

*Proof.* Because  $\mathcal{A}$  is algebraic, each output element is a known combination of the elements it received. The external  $G_1$  inputs are the KZG powers  $g_1^{\tau^i}$  for  $0 \leq i \leq n$  and  $C_\phi$ , itself such a combination; moreover  $h = g_1$ . The remaining inputs are the  $crs_z$ -elements of the CP-SNARK; by  $\tau$ -algebraic compatibility (Definition 5, assumption (iii) of Theorem 1) the reduction holds, for each  $G_1$ -component  $Z_j$  of  $crs_z$ , a known polynomial  $f_j$  of degree at most  $d_{\text{aux}} \leq n$  with  $Z_j = g_1^{f_j(\tau)}$ —without knowing  $\tau$ —and produces a joint distribution matching the real one. Hence every  $G_1$ -element the adversary receives has a known polynomial representation in  $\tau$  of degree at most  $n$ , so the AGM representations yield known polynomials  $A, B, P$ , each of degree at most  $n$ , such that

$$C_{S_R} = g_1^{A(\tau)}, \quad \pi_{S_R} = g_1^{B(\tau)}, \quad \pi_\alpha = g_1^{P(\tau)}.$$

*Part (a).* Taking the discrete logarithm of the first check to the base  $e(g_1, g_2)$ ,

$$\phi(\tau) - A(\tau) = B(\tau)V_{S_R}(\tau).$$

Set the known polynomial

$$D(T) = \phi(T) - A(T) - B(T)V_{S_R}(T),$$

whose degree is at most  $n + R$ . If  $D \neq 0$ , the accepting check forces  $\tau$  to be one of its roots, and we turn this into an  $n$ -DLOG solver. The reduction takes an  $n$ -DLOG challenge  $(\{g_1^{\tau^i}\}_{i=0}^n, \{g_2^{\tau^i}\}_{i=0}^R)$  (Definition 4), embeds it verbatim as the KZG CRS, generates  $crs_z$  itself as just described, and runs  $\mathcal{A}$ . From the AGM representations it holds  $A$  and  $B$  as explicit polynomials, so it forms  $D$  and *factors it over  $\mathbb{F}_p$  in the clear*; among the at most  $n + R$  roots it returns the  $r$  with  $g_1^r = g_1^\tau$ , reading  $g_1^\tau$  off the CRS. Factoring uses no group operations, and the reduction never forms a group element of degree exceeding  $n$ ; so the challenge need only supply the powers already present in the KZG CRS, and  $n$ -DLOG—not  $(n+R)$ -DLOG—suffices despite  $\deg D \leq n + R$ . Whenever  $\mathcal{A}$  produces an accepting transcript with  $D \neq 0$  the reduction outputs  $\tau$ , so that event has probability at most  $\epsilon_{n\text{-dl}}$ , which we fold into  $\epsilon_{\text{KZG}}$ . Hence  $D \equiv 0$ , and  $V_{S_R} \mid (\phi - A)$ . Put  $\hat{\phi}_{S_R} := A$ ; then  $C_{S_R} = g_1^{\hat{\phi}_{S_R}(\tau)}$ , and  $V_{S_R}(i) = 0$  for  $i \in S_R$  gives  $\hat{\phi}_{S_R}(i) = \phi(i)$ .

*Part (b).* Let  $y$  satisfy  $U_\alpha = g_1^y$  and be known to the reduction. Using  $C_{S_R} = g_1^{A(\tau)}$  from part (a), the second check reads  $A(\tau) - y = P(\tau)(\tau - \alpha)$ . Define

$$E(T) = A(T) - y - P(T)(T - \alpha), \quad \deg E \leq n + 1.$$

The same reduction rules out  $E \neq 0$ : here  $\deg E \leq n + 1$ , and the reduction factors  $E$  in the clear and returns its root equal to  $\tau$  exactly as in part (a), again needing only the  $n$ -DLOG powers. Whenever  $\mathcal{A}$  accepts with  $E \neq 0$  it solves  $n$ -DLOG, an event of probability at most  $\epsilon_{n\text{-dl}}$ , folded into  $\epsilon_{\text{KZG}}$ . Thus  $(T - \alpha) \mid (A(T) - y)$ ; evaluating at  $T = \alpha$  yields  $\hat{\phi}_{S_R}(\alpha) = A(\alpha) = y$ .

Both parts bound their bad event by a single computational reduction to  $n$ -DLOG, so  $\epsilon_{\text{KZG}}$  is the aggregate advantage of the constant number of such reductions and  $\epsilon_{\text{KZG}} \leq c\epsilon_{n\text{-dl}}$ . This is distinct from—and does not absorb—the Schwartz–Zippel sampling loss over the Fiat–Shamir challenges  $\alpha$  and  $S_R$ , which is a statistical term (the  $q_P n/(p - R - 2)$  and  $q_S((\beta + 1)/2\beta)^R$  contributions in Theorem 1) and enters the soundness bound separately, not through this lemma.  $\square$

*Remark 1.* The opening check *alone* does not pin  $U_\alpha$  to  $g_1^{\hat{\phi}_{S_R}(\alpha)}$ : an algebraic prover may output  $U_\alpha = g_1^{\hat{\phi}_{S_R}(\tau) + (\tau - \alpha)r(\tau)}$  for any  $r$  and still open it. Part (b) becomes meaningful only once an *explicit* scalar  $y$  with  $U_\alpha = g_1^y$  is available. In the construction this scalar is delivered by the CPLink equation  $U_\alpha = g_1^{u_\alpha}$ , which is therefore also essential to soundness, not only a zero-knowledge device.

#### A.4 Correctness

**Lemma 3 (Correctness).** *The construction of Figure 1 satisfies Definition 1.*

*Proof.* Fix  $\phi$  of degree at most  $\ell$  and run **Enc** honestly. The key check holds since  $\text{vk} = h^{\text{sk}}$  by construction. The blinded polynomial  $\phi_{S_R} = \phi_{S_R} + (t_1 + t_2 X)V_{S_R}$  has degree at most  $R + 1$  (as  $\deg \phi_{S_R} \leq R - 1$  and  $\deg V_{S_R} = R$ ) and, since



$V_{S_R}$  vanishes on  $S_R$ , satisfies  $\tilde{\phi}_{S_R}(i) = \phi_{S_R}(i) = \phi(i)$  for  $i \in S_R$ . Thus  $\phi - \tilde{\phi}_{S_R}$  vanishes on  $S_R$ , so  $Q := (\phi - \tilde{\phi}_{S_R})/V_{S_R}$  is a polynomial (degree  $\leq \max(\ell, R+1) - R \leq n$ ) and  $\pi_{S_R} = g_1^{Q(\tau)}$  gives the subset identity  $\phi(\tau) - \tilde{\phi}_{S_R}(\tau) = Q(\tau)V_{S_R}(\tau)$ , i.e.  $e(C_\phi/C_{S_R}, g_2) = e(\pi_{S_R}, g_2^{V_{S_R}(\tau)})$ ; likewise, with  $u_\alpha = \tilde{\phi}_{S_R}(\alpha)$  the quotient  $q_\alpha = (\tilde{\phi}_{S_R} - u_\alpha)/(X - \alpha)$  is a polynomial and  $\pi_\alpha = g_1^{q_\alpha(\tau)}$  gives the opening identity  $e(C_{S_R}/U_\alpha, g_2) = e(\pi_\alpha, g_2^{\tau-\alpha})$ .

For the CP-SNARK relation set  $x_i = \phi(i)$  ( $i \in S_R$ ) and  $x_i = \tilde{\phi}_{S_R}(i)$  ( $i \in \{-1, -2\}$ ); then  $ct_i = x_i + H(\text{sk}, i)$  on  $S'_R$ , and as  $\deg \tilde{\phi}_{S_R} \leq R+1 = |S'_R| - 1$ , Lagrange interpolation over the  $R+2$  points of  $S'_R$  is exact, giving  $\sum_{i \in S'_R} x_i L_{i, S'_R}(\alpha) = \tilde{\phi}_{S_R}(\alpha) = u_\alpha$  (using  $\tilde{\phi}_{S_R}(i) = \phi(i) = x_i$  on  $S_R$ ). With  $U_\alpha = g_1^{u_\alpha}$  and  $\text{vk} = h^{\text{sk}}$  the witness  $(u_\alpha, \text{sk}, \{x_i\})$  satisfies  $R_{\text{mask}}$  and both CPLink relations, so  $\pi_Z$  verifies by completeness and all three checks of  $\text{Ver}_{ct}$  accept. Finally, honest unmasking gives  $\phi'(i) = ct_i - H(\text{sk}, i) = \phi(i)$  on  $[m]$ , the exact codeword, so RS.Det reports no error and Dec outputs  $F(\phi)$ .  $\square$

## A.5 Soundness

**Lemma 4 (Soundness).** *Under the assumptions of Theorem 1, every PPT algebraic adversary with grinding budgets  $(q_S, q_P)$  has soundness advantage (Definition 2) at most*

$$\epsilon_{\text{KZG}} + \epsilon_{\text{CP}}^{\text{snd}} + q_P \frac{n}{p - R - 2} + q_S \left( \frac{\beta + 1}{2\beta} \right)^R.$$

*Proof.* Let  $\mathcal{A}$  play the game of Definition 2, outputting  $(\text{sk}^*, \text{vk}, ct, \pi)$  with  $\pi = (C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha, \pi_Z)$ . We bound the probability of the event  $\text{Win}$  that  $\text{Ver}_{ct} = 1$ ,  $\text{Ver}_{key}(\text{vk}, \text{sk}^*) = 1$ , and  $y := \text{Dec}(F, \text{sk}^*, ct) \neq F(\phi)$ . The verifier recomputes  $S_R$  and  $\alpha \notin S'_R$  from the fixed transcript.

**Step S1 (CP-SNARK extraction).** Running the knowledge extractors of the core CP-SNARK and of the CPLink instance yields a witness  $(\text{sk}, u_\alpha, \{x_i\}_{i \in S'_R})$ ; except with probability  $\epsilon_{\text{CP}}^{\text{snd}} := \epsilon_{\text{ccSNARK}}^{\text{snd}} + \epsilon_{\text{CPLink}}^{\text{snd}}$  it satisfies  $R_{\text{mask}}$  and the two linking equations, i.e.

$$ct_i = x_i + H(\text{sk}, i) \ (i \in S'_R), \quad u_\alpha = \sum_{i \in S'_R} x_i L_{i, S'_R}(\alpha), \quad U_\alpha = g_1^{u_\alpha}, \quad \text{vk} = h^{\text{sk}}. \quad (1)$$

**Step S2 (the extracted key is the decryption key).** By construction (Gen)  $h$  is a nontrivial element of the prime-order group  $G_1$ , hence a generator, so  $\log_h(\cdot)$  is a bijection. From  $\text{Ver}_{key}(\text{vk}, \text{sk}^*) = 1$  we have  $\text{vk} = h^{\text{sk}^*}$ , and from (1)  $\text{vk} = h^{\text{sk}}$ ; therefore  $\text{sk} = \text{sk}^*$ . Thus the key extracted from  $\pi_Z$  is exactly the key that Dec uses to unmask, and  $H(\text{sk}, \cdot) = H(\text{sk}^*, \cdot)$ .

**Step S3 (KZG extraction).** Apply Lemma 2. Except with probability  $\epsilon_{\text{KZG}}$ , there is  $\hat{\phi}_{S_R}$  of degree at most  $n$  with  $\hat{\phi}_{S_R}(i) = \phi(i)$  for  $i \in S_R$  (part a); and,

taking the explicit scalar  $y := u_\alpha$  from (1) (for which  $U_\alpha = g_1^{u_\alpha}$  holds, so the hypothesis of part b is met),  $\widehat{\phi}_{S_R}(\alpha) = u_\alpha$  (part b).

**Step S4 (interpolation and Schwartz–Zippel).** Let  $\psi(X) := \sum_{i \in S'_R} x_i L_{i, S'_R}(X)$ , so  $\deg \psi \leq R + 1$  and, by (1),  $\psi(\alpha) = u_\alpha$ . Combining with Step S3,  $\psi(\alpha) = \widehat{\phi}_{S_R}(\alpha)$ . Both  $\psi$  and  $\widehat{\phi}_{S_R}$  are fixed before  $\alpha$  is derived: although the proof  $\pi_Z$  from which  $\{x_i\}$  are extracted is produced after  $\alpha$ , the masking equation in (1) pins each  $x_i = ct_i - H(\text{sk}, i)$  to a value determined by the pre- $\alpha$  transcript and  $\text{sk}$ , so  $\psi = \sum_i x_i L_{i, S'_R}$  does not depend on  $\alpha$ ; and  $\widehat{\phi}_{S_R}$  is a function of  $C_{S_R}$ , which is an input to  $\alpha = \text{DerivePoint}_H(\dots, C_{S_R}, \pi_{S_R}, S_R)$  and hence also fixed beforehand. Suppose  $\psi \neq \widehat{\phi}_{S_R}$ ; then  $\psi - \widehat{\phi}_{S_R}$  is a nonzero polynomial of degree at most  $n$ , with at most  $n$  roots in  $\mathbb{F}_p$ . Each invocation of  $\text{DerivePoint}_H$  returns a value uniform on  $\mathbb{F}_p \setminus S'_R$  (size  $p - R - 2$ ), so a single derivation lands on a root with probability at most  $n/(p - R - 2)$ . Since  $\mathcal{A}$  obtains at most  $q_P$  candidate point derivations, a union bound gives  $\Pr[\psi(\alpha) = \widehat{\phi}_{S_R}(\alpha) \wedge \psi \neq \widehat{\phi}_{S_R}] \leq q_P n/(p - R - 2)$ . Except with this probability,  $\psi = \widehat{\phi}_{S_R}$  as polynomials, whence for  $i \in S_R$

$$x_i = \psi(i) = \widehat{\phi}_{S_R}(i) = \phi(i), \quad \text{i.e.} \quad ct_i - H(\text{sk}, i) = \phi(i).$$

**Step S5 (sampling to decoding).** Let  $c_i := ct_i - H(\text{sk}, i)$  for  $i \in [m]$ ; this is precisely the word  $\text{Dec}$  un.masks (Step S2). Let  $C := \{i \in [m] : c_i \neq \phi(i)\}$  be its set of disagreements with the codeword  $(\phi(0), \dots, \phi(m))$ . Step S4 shows  $S_R \cap C = \emptyset$ .

- If  $|C| \leq t = \lfloor (m - \ell)/2 \rfloor$ , then  $c$  lies within the decoding radius; by RS correctness  $\text{RS.Det}$  and (if needed)  $\text{RS.Dec}$  reconstruct  $\phi$  on  $[\ell]$ , so  $y = F(\phi)$  and  $\text{Win}$  fails.
- If  $|C| > t$ , a direct counting bound applies over the  $m + 1 = \lceil \beta(\ell + 1) \rceil$  codeword positions  $[m]$  with correction radius  $t = \lfloor (m - \ell)/2 \rfloor$  and corruption set  $C$ : the fraction of positions avoiding  $C$  is  $(m + 1 - |C|)/(m + 1) \leq (m + 1 - t)/(m + 1) \leq (\beta + 1)/2\beta$ , so a uniform  $R$ -subset of  $[m]$  avoids  $C$  with probability at most  $((\beta + 1)/2\beta)^R$  (sampling without replacement only lowers this). In the ROM,  $S_R = \text{DeriveSubset}_H(\dots)$  is computed from random-oracle outputs at the subset-derivation inputs  $(C_\phi, \text{vk}, ct, m, R, j)$ , which—under the injective, domain-separated input encoding of  $H$ —are disjoint from the masking inputs  $(\text{sk}, i)$  whose outputs determine  $c$  and hence  $C$ ; thus, conditioned on any fixed transcript,  $S_R$  is a uniform  $R$ -subset independent of  $C$ . An adversary fixes at most  $q_S$  candidate transcripts, so  $\Pr[S_R \cap C = \emptyset \wedge |C| > t] \leq q_S ((\beta + 1)/2\beta)^R$ .

Hence the event “Win with  $|C| > t$ ” (an accepting transcript whose proven  $S_R \cap C = \emptyset$  coincides with an undetected large corruption) has probability at most  $q_S ((\beta + 1)/2\beta)^R$ , and the event “Win with  $|C| \leq t$ ” is impossible.

**Collecting.** Across S1–S5,

$$\Pr[\text{Win}] \leq \epsilon_{\text{CP}}^{\text{snd}} + \epsilon_{\text{KZG}} + q_P \frac{n}{p - R - 2} + q_S \left( \frac{\beta + 1}{2\beta} \right)^R,$$

which is the claimed bound.  $\square$

### A.6 Extending soundness to an untrusted committer

Theorem 1 fixes an honestly generated degree- $\ell$  commitment (Definition 2), as in prior VECK schemes [5, 7]. When the committer is itself untrusted—most naturally when the same party commits and runs **Enc**—a single one-time check removes even this assumption, at the cost of one CRS element and no dependence on the file length.

**Proposition 1 (Degree enforcement for an untrusted committer).** *Augment the CRS with  $g_2^{\tau^{n-\ell}}$ . Before the exchange, the committer publishes  $\pi_{\text{deg}} := g_1^{\tau^{n-\ell}\phi(\tau)}$  and the buyer checks*

$$e(C_\phi, g_2^{\tau^{n-\ell}}) = e(\pi_{\text{deg}}, g_2).$$

*Under the  $n$ -DLOG assumption of Theorem 1(i), any  $C_\phi$  accepted by this check is a commitment to a polynomial of degree at most  $\ell$ , except with probability  $\epsilon_{\text{deg}}$  of the same order as  $\epsilon_{\text{KZG}}$ . Consequently the soundness bound of Theorem 1 holds verbatim, with the added term  $\epsilon_{\text{deg}}$ , for an adversarially chosen  $C_\phi$  that passes the check.*

*Proof (Proof sketch).* An honest committer can form  $\pi_{\text{deg}}$  only when  $\tau^{n-\ell}\phi(\tau)$  has degree at most  $n$ , i.e.  $\deg \phi \leq \ell$ , since the CRS exposes  $g_1$ -powers only up to  $\tau^n$ . For an algebraic committer, extract  $C_\phi = g_1^{a(\tau)}$  and  $\pi_{\text{deg}} = g_1^{b(\tau)}$  with known polynomials  $a, b$  of degree at most  $n$ . The accepted pairing check means  $a(\tau)\tau^{n-\ell} = b(\tau)$ . If  $a$  had a monomial of degree above  $\ell$ , then  $a(X)X^{n-\ell}$  would have a monomial of degree above  $n$ , which the degree- $\leq n$  polynomial  $b(X)$  cannot cancel, so  $D(X) := a(X)X^{n-\ell} - b(X) \not\equiv 0$ ; yet  $D(\tau) = 0$ , and recovering the roots of the known nonzero  $D$  yields  $\tau$ , contradicting  $n$ -DLOG. Hence  $\deg a \leq \ell$  and the committed word lies in the code the decoder assumes. The remaining soundness argument is unchanged, as it already treats  $(\mathbf{vk}, \mathbf{sk}, ct, \pi)$  as fully adversarial.

### A.7 Zero-knowledge

**Lemma 5 (Computational zero-knowledge).** *Under the assumptions of Theorem 1, the construction satisfies Definition 3 with advantage at most  $\epsilon_{\text{CP}}^{\text{zk}} + \epsilon_{\text{DL}}^{G_1} + q_P/p$ .*

*Proof. The simulator.* As is standard for non-interactive zero-knowledge, we prove the property in the model where the simulator also simulates the SNARK setup. Let  $(S_{\text{kg}}^{\text{CP}}, S_{\text{prv}}^{\text{CP}})$  be the simulator guaranteed by the composable zero-knowledge of the CP-SNARK (Theorem 1(ii)):  $S_{\text{kg}}^{\text{CP}}$  produces a reference string  $\text{crs}_z$  together with a simulation state  $\text{st}_{\text{CP}}$ , and  $S_{\text{prv}}^{\text{CP}}(\text{st}_{\text{CP}}, \cdot)$  produces accepting proofs for arbitrary well-formed statements, with indistinguishability from honest proofs on true statements; nothing below uses the internal form of  $\text{st}_{\text{CP}}$ .

$\text{Sim}(\widetilde{pp}, \text{st}_{\text{CP}}, F, C_\phi)$  receives the simulated CP reference string and state produced in Definition 3 and controls the random oracle  $H$ ; the external KZG  $\text{crs}$  (hence  $\tau$ ) is given and is *not* simulated.  $\text{Sim}$  proceeds as follows. (1) Sample  $\text{sk} \xleftarrow{\$} \mathbb{F}_p$  and set  $\text{vk} = h^{\text{sk}}$ . (2) Sample  $ct_i \xleftarrow{\$} \mathbb{F}_p$  for all  $i \in [m] \cup \{-1, -2\}$ . (3) Derive  $S_R = \text{DeriveSubset}_H(C_\phi, \text{vk}, \{ct_i\}_{i \in [m]}, m, R)$  (which depends only on the ciphertext, not on  $C_{S_R}$ ); sample  $y_1 \xleftarrow{\$} \mathbb{F}_p$  and set  $\pi_{S_R} = g_1^{y_1}$ ,  $C_{S_R} = C_\phi \cdot g_1^{-y_1 V_{S_R}(\tau)}$ , where  $g_1^{V_{S_R}(\tau)} = \prod_i (g_1^{\tau})^{v_i}$  (for  $V_{S_R}(X) = \sum_i v_i X^i$ ) is a public combination of CRS elements, well-defined since  $\deg V_{S_R} = R \leq n$ ; then derive  $\alpha = \text{DerivePoint}_H(\dots, C_{S_R}, \pi_{S_R}, S_R)$ . (4) Sample  $y_2 \xleftarrow{\$} \mathbb{F}_p$  and set  $\pi_\alpha = g_1^{y_2}$ ,  $U_\alpha = C_{S_R} \cdot g_1^{-y_2(\tau-\alpha)}$  (again a public CRS combination). (5) Set  $\pi_Z \leftarrow S_{\text{prv}}^{\text{CP}}(\text{st}_{\text{CP}}, \text{stmt})$  for the public statement  $(h, S'_R, \{ct_i\}_{i \in S'_R}, \{L_{i, S'_R}(\alpha)\}_{i \in S'_R}, U_\alpha, \text{vk})$ . (6) Output  $(\text{vk}, ct, (C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha, \pi_Z))$ .  $\text{Sim}$  uses only  $C_\phi$ , the public CRS, and fresh randomness; it never uses  $\phi$ ,  $\tau$ , or the KZG trapdoor.

We bound the distinguishing advantage through hybrids  $H_0, \dots, H_4$ , with  $H_0$  the real view (including  $pp$ ) and  $H_4$  the simulated view (including  $\widetilde{pp}$ ). Throughout we condition on  $\tau \neq \alpha$ ; since  $\tau$  is uniform over  $\mathbb{F}_p$  (the external CRS trapdoor) and independent of the at most  $q_P$  point-oracle outputs,  $\Pr[\tau = \alpha] \leq q_P/p$ . We also use  $\alpha \notin S'_R$  (so  $V_{S_R}(\alpha) \neq 0$ ), which always holds by construction.

$H_0 \rightarrow H_1$  (*simulate  $\pi_Z$* ). Replace the honest SNARK reference string  $\text{crs}_z$  by a simulated one (produced by  $S_{\text{kg}}^{\text{CP}}$ , with state  $\text{st}_{\text{CP}}$ ) and  $\pi_Z$  by  $S_{\text{prv}}^{\text{CP}}$  on the same statement; composable zero-knowledge of the CP-SNARK bounds both changes at once. At  $H_0$  the statement is true (it has the honest witness), so composable zero-knowledge applies directly and  $|\Pr[\mathcal{D}(H_1)] - \Pr[\mathcal{D}(H_0)]| \leq \epsilon_{\text{CP}}^{\text{zk}}$ . Composable zero-knowledge is invoked only here, where the statement is true. Later hybrids use arbitrary-statement simulation on both sides as required by the interface, without comparing a simulated false-statement proof to an honest proof.

$H_1 \rightarrow H_2$  (*uniform ciphertext*). Replace every  $ct_i$  ( $i \in [m] \cup \{-1, -2\}$ ) by a uniform field element and derive  $S_R$  from  $\{ct_i\}_{i \in [m]}$  and  $\alpha$  from the full resulting transcript; the KZG components  $C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha$  remain the honest ones (computed from  $\widetilde{\phi}_{S_R}$  and the blinders  $t_1, t_2$  for the new  $S_R$ ). Let  $Q$  be the event that  $\mathcal{D}$  never queries  $H$  with first coordinate  $\text{sk}$  (at any second input). Conditioned on  $Q$ , the pads  $H(\text{sk}, i)$  for  $i \in [m] \cup \{-1, -2\}$  are uniform and independent in  $\mathcal{D}$ 's view— $\mathcal{D}$  sees only the ciphertext entries, not the simulator's own oracle evaluations—so the real  $ct_i = \phi(i) + H(\text{sk}, i)$  (and  $ct_i = \widetilde{\phi}_{S_R}(i) + H(\text{sk}, i)$  for  $i \in \{-1, -2\}$ ) are uniform, identical to  $H_2$ ; thus  $H_2 \equiv H_1$  given  $Q$ . We bound  $\Pr[\neg Q]$  by a discrete-log reduction: on challenge  $(g_1, T)$ , set  $h := g_1$ ,  $\text{vk} := T$ , and run the  $H_2$  experiment, which needs neither  $\text{sk}$  nor  $\tau$ ; whenever  $\mathcal{D}$  queries  $H(s', \cdot)$  with  $g_1^{s'} = T$ , output  $s' = \log_{g_1} T$ . Hence  $\Pr[\neg Q] \leq \epsilon_{\text{DL}}^{G_1}$ . *Crucially, after this step the ciphertext is independent of the blinders  $(t_1, t_2)$* : the auxiliary entries  $ct_{-1}, ct_{-2}$ , which in  $H_0, H_1$  carry  $\widetilde{\phi}_{S_R}(-1), \widetilde{\phi}_{S_R}(-2)$  and are therefore correlated with  $u_\alpha$  through  $(t_1, t_2)$ , are now uniform; the only view components that still depend on  $(t_1, t_2)$  are the KZG transcripts  $C_{S_R}, \pi_{S_R}, U_\alpha, \pi_\alpha$ . This is what makes the next two determinant arguments exact.

$H_2 \rightarrow H_3$  (*rewrite the opening transcript*). Replace  $(U_\alpha, \pi_\alpha)$  by  $\pi_\alpha = g_1^{y_2}$  ( $y_2$  uniform) and  $U_\alpha = C_{S_R} g_1^{-y_2(\tau-\alpha)}$ , and re-simulate  $\pi_Z$ . We claim  $H_3 \equiv H_2$  exactly. By  $H_2$  the only  $(t_1, t_2)$ -dependent view components are the KZG transcripts, so we may argue conditionally on  $C_{S_R}$ . The honest opening value is  $u_\alpha = \tilde{\phi}_{S_R}(\alpha) = \phi_{S_R}(\alpha) + bV_{S_R}(\alpha)$  with  $b = t_1 + t_2\alpha$ ; since  $(t_1, t_2) \mapsto (a, b) = (t_1 + t_2\tau, t_1 + t_2\alpha)$  has determinant  $\alpha - \tau \neq 0$ ,  $b$  is uniform given  $a$  — equivalently given  $(C_{S_R}, \pi_{S_R})$ , since both are functions of  $a$  alone ( $C_{S_R} = g_1^{\phi_{S_R}(\tau) + aV_{S_R}(\tau)}$  and, as shown in the next step,  $\pi_{S_R} = g_1^{g(\tau)-a}$ ) — and because  $V_{S_R}(\alpha) \neq 0$  (as  $\alpha \notin S'_R$ ) the value  $U_\alpha = g_1^{u_\alpha}$  is uniform given  $(C_{S_R}, \pi_{S_R})$ ; the opening equation forces  $\pi_\alpha = (C_{S_R}/U_\alpha)^{1/(\tau-\alpha)}$ . In  $H_3$ ,  $\pi_\alpha = g_1^{y_2}$  is uniform given  $C_{S_R}$  and  $U_\alpha$  is forced by the same equation. For fixed  $C_{S_R}$  the map  $U_\alpha \leftrightarrow \pi_\alpha = (C_{S_R}/U_\alpha)^{1/(\tau-\alpha)}$  is a bijection (using  $\tau \neq \alpha$ ), carrying the uniform law of  $U_\alpha$  to that of  $\pi_\alpha$ ; hence the conditional laws of  $(U_\alpha, \pi_\alpha)$  given  $C_{S_R}$  coincide in  $H_2$  and  $H_3$ , and the rest of the view is an identical function of  $(C_{S_R}, U_\alpha, \pi_\alpha)$ . Hence  $H_3 \equiv H_2$ .

$H_3 \rightarrow H_4$  (*rewrite the subset transcript*). Replace  $(C_{S_R}, \pi_{S_R})$  by  $\pi_{S_R} = g_1^{y_1}$  ( $y_1$  uniform) and  $C_{S_R} = C_\phi g_1^{-y_1 V_{S_R}(\tau)}$ ; since  $\alpha = \text{DerivePoint}_H(\dots, C_{S_R}, \pi_{S_R}, S_R)$  takes these as inputs, re-derive  $\alpha$ , refresh  $U_\alpha = C_{S_R} g_1^{-y_2(\tau-\alpha)}$ , and re-simulate  $\pi_Z$ . It suffices to show the joint law of  $(C_{S_R}, \pi_{S_R})$  is preserved, since then the induced laws of  $\alpha$ , of the refreshed  $U_\alpha$ , and of the rest of the view are unchanged. If  $\tau \notin S_R$  (so  $V_{S_R}(\tau) \neq 0$ ), the real pair has  $C_{S_R}$  uniform with  $\pi_{S_R} = (C_\phi/C_{S_R})^{1/V_{S_R}(\tau)}$  forced, while  $H_4$  has  $\pi_{S_R}$  uniform with  $C_{S_R}$  forced by the same relation; the bijection  $C_{S_R} \leftrightarrow \pi_{S_R}$  (using  $V_{S_R}(\tau) \neq 0$ ) carries one law to the other, so  $H_4 \equiv H_3$ . In the residual case  $\tau \in S_R$  we have  $V_{S_R}(\tau) = 0$ : both hybrids fix  $C_{S_R} = C_\phi$  (in  $H_3$ ,  $C_{S_R} = g_1^{\phi_{S_R}(\tau)} = g_1^{\phi(\tau)} = C_\phi$  because  $\phi_{S_R}$  interpolates  $\phi$  at  $\tau \in S_R$ ; in  $H_4$ ,  $C_{S_R} = C_\phi g_1^0 = C_\phi$ ). For  $\pi_{S_R}$ : writing  $\phi - \phi_{S_R} = V_{S_R} g$  (a polynomial identity, as  $\phi \equiv \phi_{S_R}$  on  $S_R$ ), the honest quotient is  $(\phi - \phi_{S_R})/\tilde{V}_{S_R} = g - (t_1 + t_2 X)$ , so  $\pi_{S_R} = g_1^{g(\tau)-a}$  with  $a = t_1 + t_2\tau$  uniform and unconstrained (because  $C_{S_R} = C_\phi$  regardless of  $a$ , and by  $H_2$  the ciphertext no longer constrains  $(t_1, t_2)$ ); in  $H_4$ ,  $\pi_{S_R} = g_1^{y_1}$  is uniform. So  $H_4 \equiv H_3$  in either case. After  $H_4$ , every element is computed from  $C_\phi$ , the CRS,  $S_R$ ,  $\alpha$ , and fresh  $y_1, y_2$  only—i.e.  $H_4 = \text{Sim}$ .

*Total.* Summing the transitions and the conditioning events,

$$\epsilon_{\text{zk}} \leq \epsilon_{\text{CP}}^{\text{zk}} + \underbrace{\epsilon_{\text{DL}}^{G_1}}_{H_1 \rightarrow H_2: \Pr[\neg Q]} + \underbrace{0}_{H_2 \rightarrow H_3} + \underbrace{0}_{H_3 \rightarrow H_4} + \underbrace{qP/p}_{\Pr[\tau=\alpha]} = \epsilon_{\text{CP}}^{\text{zk}} + \epsilon_{\text{DL}}^{G_1} + \frac{qP}{p}.$$

□

## A.8 Proof of Theorem 1

*Proof (Proof of Theorem 1).* Correctness is Lemma 3. Soundness is Lemma 4, whose bound  $\epsilon_{\text{KZG}} + \epsilon_{\text{CP}}^{\text{snd}} + qPn/(p - R - 2) + q_S((\beta + 1)/2\beta)^R$  is exactly the soundness advantage stated in the theorem; under the parameter choice  $q_S((\beta + 1)/2\beta)^R \leq 2^{-\lambda}$  the sampling term is negligible in  $\lambda$ . Zero-knowledge is Lemma 5.

All three hold in the AGM and ROM under the stated assumptions, so the construction is a secure VECK in the sense of Definitions 1–3.  $\square$